



Sphalerite Chemistry and Carbonate (C, O) Isotope Zonation in the Nayongzhi Mississippi Valley-Type Deposit, Southwest China

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Abstract

Metal zonation patterns are common in Mississippi Valley-type (MVT) deposits but have received comparatively little attention. The recognition of such patterns can help to fingerprint fluid flow paths and provide valuable vectors for near-mine exploration. The Nayongzhi deposit (30.1 million tonnes [Mt] at 6.57% Zn and 0.91% Pb) exhibits clear lateral and vertical zonation patterns in ore types and metal ratios, making it an ideal case study to explore the behavior of other geochemical zonation markers. In this contribution, we report zonation patterns observed in both carbonate C-O isotope compositions and the trace element chemistry of sphalerite. Ore-related dolomite shows systematically higher $\delta^{13}\text{C}_{\text{V-PDB}}$ (Vienna-Pee Dee Belemnite) and $\delta^{18}\text{O}_{\text{V-SMOW}}$ (Vienna-standard mean ocean water) values with increasing distance to the feeder structure, probably reflecting a decrease in the fluid-rock ratios. In addition, laser ablation-inductively coupled plasma-mass spectrometry (LA-ICP-MS) analysis of sphalerite revealed systematic spatial and temporal variations in trace element contents. The contents of Mn, Fe, Cu, Ga, Ge, and In decrease with distance from the feeder fault, whereas the contents of Cd increase. Manganese, Fe, Ge, and Tl are also concentrated in early sphalerite, whereas Ga and Cd are more abundant in later sphalerite. Such variations appear to be predominantly governed by changes in physicochemical parameters, such as temperature, f_{S_2} , and fluid composition. Finally, multivariate regression analysis identified three footprint elements in sphalerite—Mn, Ga, and In—whose combined compositional variations are particularly informative concerning the relative distances of samples to the feeder fault. The application of the resultant sphalerite proximator to another structurally controlled MVT deposit effectively traced fluid flow paths and identified the relative distances of samples to the fluid influx center. Consequently, our study highlights the potential of sphalerite trace element chemistry, together with C-O isotopes, to serve as a novel exploration tool in MVT deposits. However, our results also demonstrate that the accurate petrologic identification of paragenetic mineral generations will generally be essential for reliably vectoring toward fluid influx centers.

Introduction

Mississippi Valley-type (MVT) deposits are important sources of lead and zinc and also often contain economic amounts of Ag, Co, Cd, Ga, and Ge (Leach et al., 2005; Wilkinson, 2014; Marsh et al., 2016). They are normally hosted in specific carbonate units and often display a structural association with extensional faults (Leach et al., 2005; Paradis et al., 2007). Some of the deposits exhibit mineralogical and chemical zonation due to the dominantly unidirectional influx of warm (100°–250°C), metal-rich fluids from specific basement-rooted faults (Mavrogenes et al., 1992; Hitzman and Beaty, 1996; Fusciardi et al., 2003; Reid et al., 2013; Frenzel et al., 2019). Well-known examples of such zoned deposits occur in the southeast Missouri district, United States, and the Polaris district, Canada (Mavrogenes et al., 1992; Reid et al., 2013; Cavender et al., 2016). The proximal mineralization commonly contains elevated levels of Zn, Pb, and Fe or Co, Ni, and Cu that decrease outward and upward. Zoning patterns of Cu, Co, Ni, As, and Zn/Pb ratios are also observed in some deposits of the Irish Midlands, such as Lisheen, Silvermines,

and Tynagh (Hitzman and Beaty, 1996; Fusciardi et al., 2003; Torremans et al., 2018), although some geologists classify Irish-type deposits separately (e.g., Hitzman and Large, 1986; Wilkinson, 2014).

The recognition of such zonation patterns is critical to fingerprint fluid flow paths and guide mineral exploration (Hagni, 2006; Reid et al., 2013; Cavender et al., 2016; Torremans et al., 2018). This process is labor intensive and costly, however, so the development of vectoring tools to footprint mineralized centers through limited numbers of drill core analyses is of significant interest. Hydrothermal minerals, such as dolomite, pyrite, sphalerite, and galena, may be of particular use as exploration tools due to spatial variations in their trace element and isotopic compositions (e.g., Barker et al., 2013; Dehnavi et al., 2018; Navarro-Ciurana et al., 2023).

China hosts abundant MVT deposits, most of which are concentrated along the margins of the Yangtze block. Specifically, the Sichuan-Yunnan-Guizhou MVT province is one of the largest in the world (Leach and Song, 2019). In the Sichuan-Yunnan-Guizhou triangle region, more than 450 MVT Zn-Pb deposits and occurrences have been identified, includ-

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ing eight of the top 15 Chinese MVT deposits, namely Huize, Maoping, Zhugongtang, Daiangzi, Tianbaoshan, Maozu, Leihong, and Nayongzhi. The triangle contains total resources exceeding 2,000 Mt of ore with grades of 8 to 35% of combined Pb and Zn, making it one of the most significant Pb and Zn provinces in China and the world (Han et al., 2023). Zinc and Pb mineralization in this triangle shows clear structural control. The mineral and chemical zonation patterns of individual deposits, however, are generally poorly documented. Some studies investigate zonations in hydrothermal carbonate minerals (dolomite and calcite) and sulfides in terms of the large-scale spatial distribution of these minerals or in terms of small-scale variations at the sample level (e.g., Yang et al., 2013; Wei, A.Y., et al., 2015; Wei, C., et al., 2021a), but little

is known about the spatial variations in ore mineral compositions and stable isotopes (e.g., C and O).

The Nayongzhi MVT deposit (26°45'N, 105°38'E) is located in the eastern margin of the Sichuan-Yunnan-Guizhou MVT province (Fig. 1A). Based on 2018 data, the deposit contains 30.1 Mt of ore (production to date plus remaining resources) with grades of 6.57% Zn and 0.91% Pb, at a cutoff grade of 0.5% Pb + Zn (Wei et al., 2018; Zhou et al., 2018). Exploration is ongoing, and resources may increase (Wang et al., 2020). Wei et al. (2024a) reported that the deposit is laterally zoned in terms of textural style, metal grades, and Pb isotopes. Proximal mineralization commonly contains open-space-filling ores with elevated Zn and Pb grades, whereas the Zn/Pb ratios increase outward. Because of the availability of

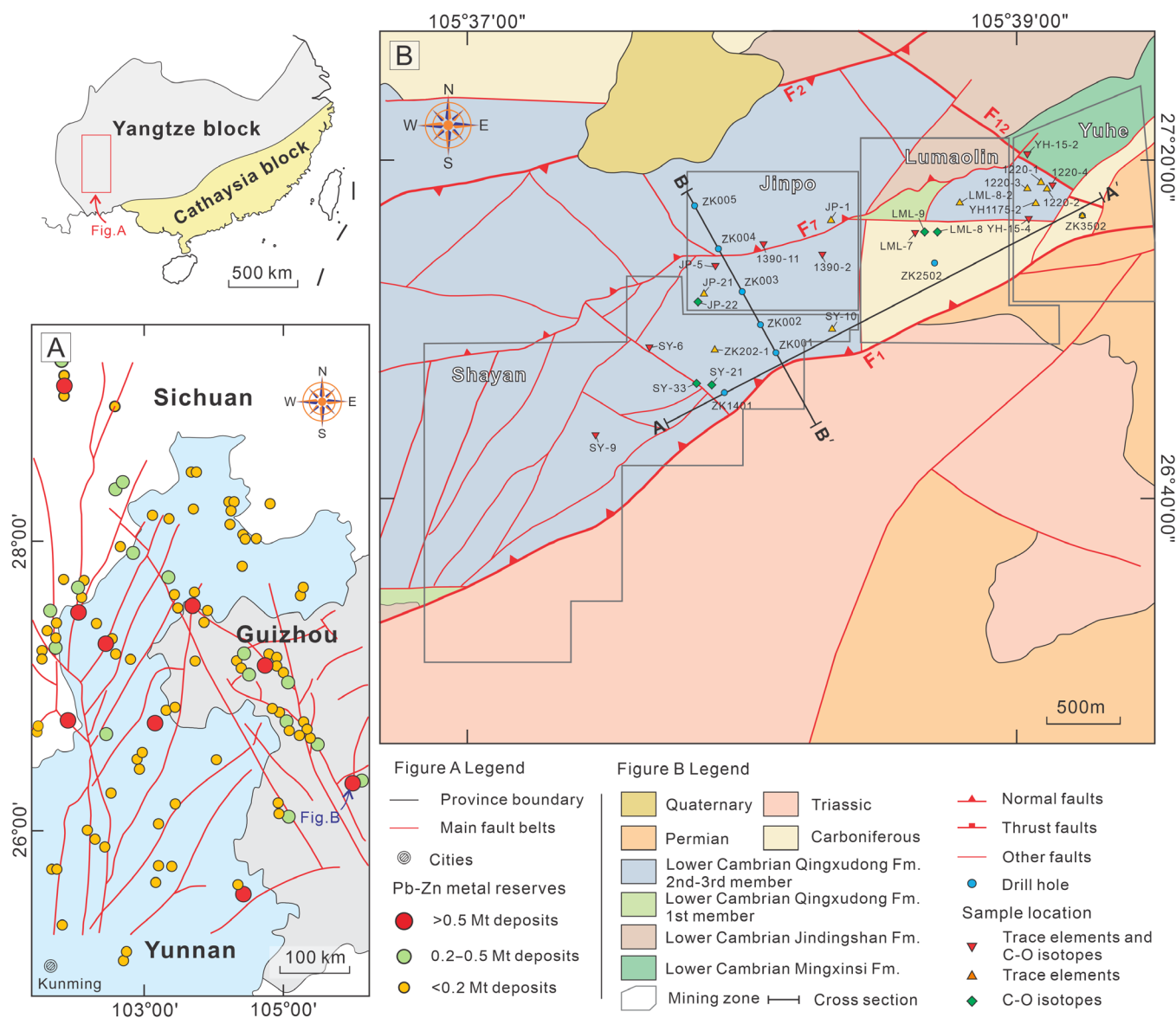


Fig. 1. Regional and local geology of the Nayongzhi deposit. (A) Geologic map of the Sichuan-Yunnan-Guizhou triangle, China, compiled from Wei et al. (2024a), including the locations of the Mississippi Valley-type (MVT) deposits and regional structures. (B) Geologic map of the Nayongzhi deposit revealing the stratigraphy and structural framework (modified from Wei et al., 2021a). Sampling locations are also provided, with sample details shown in Appendix Table A1. Abbreviations: Fm. = Formation, Mt = million tonnes.

numerous well-documented samples from historical mining operations and drill cores collected during the past eight years of geologic campaigns, the deposit was chosen as a case study to investigate zonation patterns in MVT deposits.

Previous wet chemistry studies showed that sphalerite from Nayongzhi is relatively enriched in Ga, Ge, and Cd and depleted in Mn, Fe, Co, and In, indicating the ores formed under low-temperature conditions (<155°C; Wei et al., 2018). Trace element variations across different paragenetic stages have been reported in MVT deposits (e.g., Torró et al., 2023). It remains unclear whether sphalerite chemistry in Nayongzhi exhibits systematic spatial and temporal variations, and if so, which factors controlled them. Previous carbon and oxygen isotope analyses of Nayongzhi calcite showed that the $\delta^{13}\text{C}_{\text{V-PDB}}$ (Vienna-Pee Dee Belemnite) values are between -4.1 and $+0.5\text{‰}$, suggesting carbon was sourced from the host carbonate rocks due to carbonate dissolution. The $\delta^{18}\text{O}_{\text{V-SMOW}}$ (Vienna-standard mean ocean water) values are between 12.4 and 14.1‰ , which have been interpreted as recording a mixed signal from metamorphic fluids and the original carbonate rocks (Zhou et al., 2018). It is unclear whether the C-O isotope compositions exhibit any progressive changes away from the feeder structure as observed in other MVT deposits (Navarro-Ciurana et al., 2023).

In this contribution, the temporal-spatial variations of sphalerite mineral chemistry and ore-related dolomite C-O isotope compositions in the Nayongzhi MVT deposit are first documented, based on representative samples taken along both lateral and vertical transects across the deposit (Fig. 1B). This data set, complemented by data from a previous fluid inclusion study (Wei et al., 2024a), is then used to (1) identify the key controlling factors of the patterns and (2) develop a sphalerite chemistry vector to aid exploration within the Sichuan-Yunnan-Guizhou MVT province. Finally, the general implications of our observations for MVT deposits elsewhere are considered.

Geologic Setting

The world-class Sichuan-Yunnan-Guizhou MVT province is located on the southwestern margin of the Yangtze block (Fig. 1), adjacent to the suture zone between the Indian and Eurasian blocks. The province covers an area of $\sim 170,000\text{ km}^2$ across southwestern Sichuan, northeastern Yunnan, and northwestern Guizhou (Fig. 1A; Zhang et al., 2015). In this region, the basement is composed of the late Paleoproterozoic to early Neoproterozoic Dongchuan and Meso- to early Neoproterozoic Kunyang (also called Huili) groups, which dominantly comprise siltstone, slate, sandstone, and dolostone interbedded with tuffaceous rocks and are tightly folded and weakly metamorphosed (Zhao et al., 2019). From the late Neoproterozoic to early Permian, the Yangtze block was located at low paleolatitudes, mostly surrounded by passive continental margins (Metcalf, 2013), resulting in the formation of abundant carbonate platform sequences and the accumulation of evaporites. During the late Paleozoic (480–410 Ma), the Yangtze block was influenced by the Wuyi-Yunkai orogen (also called Caledonian orogen in China) as a result of the collision between the eastern Yangtze and Cathaysia basements (Li et al., 2010; Wang et al., 2013). During this period, successive submarine sedimentary sequences of sandstone,

shale, dolostone, and limestone were deposited within the Sichuan-Yunnan-Guizhou triangle. The Emeishan flood basalt erupted at $\sim 260\text{ Ma}$ and covered an area of $\sim 250,000\text{ km}^2$, with varying thicknesses from several hundred meters to 5 km (e.g., Xu et al., 2001). From the Middle to Late Triassic (230–200 Ma), the southwestern Yangtze block was affected by the closure of the Paleo-Tethys Ocean and subsequently the amalgamation of the Indochina and South China blocks (Indosinian orogeny; Cai and Zhang, 2009; Faure et al., 2014). This resulted in the development of several thrust belts and foreland basins around the studied region. Subsequently, the Sichuan-Yunnan-Guizhou triangle transitioned to terrigenous sedimentation and was subject to continuous intracontinental deformation during the Yanshanian (180–110 Ma) and probably the Himalayan-Tibetan (65 Ma–Present) orogenies (Burchfiel and Chen, 2012; Yan et al., 2018). A series of terrigenous sandstone, conglomerate, and marl units was deposited in the region from the Jurassic to the Cenozoic.

The Sichuan-Yunnan-Guizhou triangle is bounded by three regional tectonic belts: the NS-trending Anninghe-Luzhijiang, the NE-trending Mile-Shizhong-Shuicheng, and the EW-trending Kangdian-Yiliang-Shuicheng belts (Zhang et al., 2015). These fault belts were activated and reactivated during the Wuyi-Yunkai, Indosinian, Yanshanian, and Himalayan orogenies. Secondary faults and fold-and-thrust belts also developed, trending mostly north-south to northeast and northwest throughout the region (Cai and Zhang, 2009; Burchfiel and Chen, 2012).

Economically significant Zn-Pb deposits in the Sichuan-Yunnan-Guizhou triangle are mostly hosted in the carbonate rocks of late Neoproterozoic to early Permian age (Zhang et al., 2015; Han et al., 2023). The main ore-hosting sequences are carbonate rocks of the late Ediacaran to early Cambrian and Late Devonian to early Carboniferous age, possibly due to the presence of abundant evaporites in these formations that could provide the sulfur for sulfide precipitation, though evaporites are poorly preserved in the rock record (Jin et al., 2006). Regionally, most known MVT Zn-Pb deposits are controlled by NW-trending extensional faults and the NE- and NW-trending fold-and-thrust systems (Han et al., 2023). Sulfide ores are commonly located on or near extensional faults. In some cases, the ores are associated with fold-and-thrust structures, where sulfide ores occur either as carbonate-replacement lenses parallel to bedding planes or as breccia within steep faults. Despite their close spatial association, these MVT ores generally formed later than the fold-and-thrust structures, as exemplified by the Zhugongtang deposit (Wei et al., 2021b).

Many studies have attempted to date ore formation for the MVT deposits in the region (Li et al., 2007; Yin et al., 2009; Zhang et al., 2015; Zhou et al., 2015; Xiong et al., 2018; Tang et al., 2019, and references therein). The timing of mineralization has been determined for specific MVT deposits such as Huize (Li et al., 2007; Yin et al., 2009; Cui et al., 2024) and Jinshachang (Zhang et al., 2015; Zhou et al., 2015) using multiple methods, including sphalerite Rb-Sr, carbonate and fluorite Sm-Nd, and carbonate U-Pb dating. The results indicate that most of the MVT ores formed between 225 and 192 Ma (e.g., Tang et al., 2019, and references therein; Yang et al., 2019), corresponding to the Indosinian tectonic event (Cai

and Zhang, 2009; Faure et al., 2014). However, some early Paleozoic ages have also been obtained, mostly ranging from 458 to 411 Ma (e.g., Wang et al., 2018; Xiong et al., 2018). These coincide broadly with the Caledonian orogeny (Li et al., 2010; Wang et al., 2013). Consequently, two important stages of MVT Zn-Pb ore formation appear to have occurred in the region (Xue et al., 2025).

Deposit Geology and Previous Studies

Lithostratigraphy

The Nayongzhi area contains sedimentary rocks of Cambrian and Carboniferous to Late Triassic age, unconformably overlying the Neoproterozoic basement rocks. The oldest rocks cropping out in the area are lower Cambrian silty claystone, quartz sandstone, and medium-bedded oolitic limestone of the Mingxinsi Formation (~100 m; Figs. 1B, 2A). The Jindingshan Formation (~400 m) follows the Mingxinsi Formation and contains a succession of interbedded argillaceous siltstone and quartz sandstone with minor thin-bedded limestone. The ore-hosting Qingxudong Formation (~388 m) overlies the Jindingshan Formation. This formation can be subdivided into three members from bottom to top. The first member (~130 m) consists of interbedded argillaceous and carbonaceous dolostone with an ~3-m-thick layer of sandstone and mudstone toward the upper parts. The second member, approximately 150 m thick, comprises laminated dolostone alternating with thin layers of zebra dolostone. The third member, ~108 m thick, contains fine-grained dolostone interbedded with thin-bedded argillaceous dolostone. The Qingxudong Formation is unconformably overlain by several units, including Carboniferous dolostone of the Xiangbai Formation (~20 m), Carboniferous argillaceous siltstone of the Dapu Formation (~87 m), lower Permian quartz sandstone of the Liangshan Formation (~30 m), and Lower Triassic siltstone and claystone of the Daye Formation (~60 m). Quaternary sediments cover the Cambrian and Carboniferous units in the north-central area.

Structural setting

The hydrothermal mineralization at Nayongzhi was fundamentally structurally controlled (Chen et al., 2015; Yang et al., 2018). In the deposit area, a dome structure with N30°E–N40°E-trending axis (Fig. 1B) and three main faults, namely the Xiongjiachang (F₁) thrust fault in the south, the Jichangpo (F₂) thrust fault in the north, and the F₁₂ fault in the east, are the most conspicuous structures (Fig. 1B). Specifically, the F₁ fault extends >10 km, striking N40°E–N45°E and dipping 57° to 60° toward the southeast. The F₂ fault is 6.8 km long, strikes N35°E, and dips at 60° to 70° toward the northwest. The F₁₂ fault probably originated as a NE-dipping listric thrust fault, later reactivated as an extensional fault during the Late Ordovician (Lan et al., 2019).

The zone between the three main faults is structurally complex and is segmented into smaller fault-bounded blocks by a series of NNE- and NNW-trending synthetic splay faults, many of which terminate against the main structures (Fig. 1B). The NE-trending steep F₇ fault, situated along the dome axis, displaces the F₁₂ main fault, moving the hanging wall block upward and westward by up to 50 m. Displacement transfer between the F₁ and F₂ major faults is accommodated

by minor strike-slip faults (Fig. 1B; Chen et al., 2015; Yang et al., 2018). The major displacements are thought to have occurred during the Yanshanian orogeny (Yang et al., 2018; Lan et al., 2019).

Previous work by Wei et al. (2024a) on the textural, chemical, and isotopic zonation of the deposit identified the southern segments of the F₁₂ fault as the main feeder structures for Zn-Pb mineralization in the Nayongzhi deposit.

Mineralization

The Nayongzhi sulfide ores are hosted in the first and second members of the Qingxudong Formation (Fig. 2A). The deposit is subdivided into four main mining zones, which, from northeast to southwest, are Yuhe, Lumaolin, Jinpo, and Shayan (Fig. 1B). These mining zones align with some of the key geologic features of the deposit (Wei et al., 2024a).

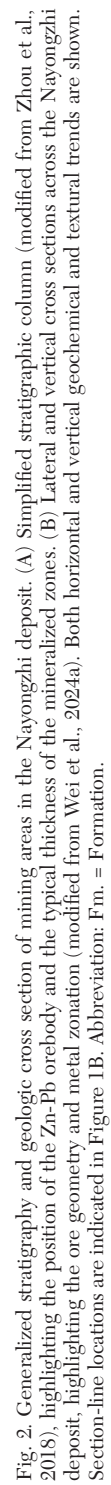
The lateral cross section through all mining zones shows two groups of sulfide orebodies categorized based on their geometry (Fig. 2B): (1) proximal, discordant orebodies filling the F₁₂ fault and cutting across the Qingxudong Formation; and (2) more distal, stacked stratabound lenses that are parallel or subparallel to the bedding of the Qingxudong Formation.

The most proximal, northeastern mineralized zone (Yuhe) exhibits both discordant and stratabound ores with a broadly tabular to irregular geometry, formed by infilling of karst-like cavities and fault fractures (Fig. 2B). The discordant ore extends vertically from 50 m above the upper boundary of the second member of the Qingxudong Formation to unknown depth. Away from the F₁₂ feeder fault toward the southwest, the orebody transitions to stratabound lenses, with a gradual decrease in cumulative thickness and ore grade (Fig. 2B).

The ores in the middle section of the deposit, including the Lumaolin and Jinpo zones, are stratabound and can be subdivided into three horizons (Fig. 2B). The lowermost horizon, located ~20 m above the base of the Qingxudong Formation, occurs as discontinuous lenses with a thickness of 10 to 20 m (Fig. 2B). The upper two horizons, located near the boundary of the first and second members of the Qingxudong Formation, have average thicknesses of 20 to 35 m and 10 to 25 m, respectively (Wang et al., 2020). Each horizon consists of different ore layers that are laterally interconnected. In the distal Shayan zone, two upper interconnected ore horizons are present with a cumulative thickness from a few centimeters to a maximum of 18.5 m (Fig. 2B). The ore horizons laterally transition into barren dolostones.

Metal and mineralogical zonation

Zonation patterns are apparent in the Nayongzhi deposit (Fig. 2; Wei et al., 2024a). With distance from the feeder fault, there are systematic changes in ore thickness, metal grades, and ore textures. Total thickness of the sulfide ores decreases toward the southwest, away from the feeder fault, from >200 m in the feeder zone to between 10 and 30 m in the most distal Shayan zone (Fig. 2B). A similar thinning toward the periphery is seen in the north to south cross section through the middle of the deposit in Figure 2B, where drill cores near the central part of the deposit (ZK002 and ZK003) intersect the greatest cumulative thicknesses of sulfide ore (up to 18.5 m) and drill cores on the periphery (ZK004, 3.3 m and ZK001, 4.7 m) show much thinner intersections.



Ore textures also show a gradual shift with increasing distance from the feeder fault (Wei et al., 2024a). Open-space-filling sulfides dominate in the Yuhe and Lumaolin zones, where the ores are massive or where they cement dolostone breccia clasts (Fig. 3A, B). Away from the feeder fault, the ores dominantly occur as infillings within dissolution cavities. Sulfide-hydrothermal dolomite assemblages are concentrated within such cavities (Fig. 3C) and, to a lesser extent, occur as small, irregular sulfide pods scattered around the dolostone breccias (Fig. 3D). Carbonate-replacement ores gradually become dominant in the Jinpo zone. This replacement of the host rocks commonly preserves dolostone "ghost textures" (Fig. 3E). The sulfides partially or completely replace clasts of fractured dolostone in these ores (Fig. 3F). In some cases, crack- and cavity-rich host rocks are also observed, which are more susceptible to mineralization owing to their higher porosity (Fig. 3G). In the distal Shayan zone, sulfides selectively replace the host dolostones (Fig. 3H) or occur as small pods associated with bioclasts or spar-filled cavities in the host rock (Fig. 3I, J). The most common mineral in the Shayan zone is replacement dolomite, accompanied by a small proportion of sulfides (Fig. 3J).

In addition to changes in ore thickness and textures, a clear metal zonation is evident at Nayongzhi. Ores in the feeder zones usually have high Zn and Pb grades that decrease laterally outward and low Zn/Pb ratios that increase in the same direction (Fig. 2; Wei et al., 2024a). Vertical zonation is also present but insignificant relative to the lateral variations, in particular for Zn and Pb grade variations (Fig. 2B).

Sphalerite, galena, and pyrite are the dominant sulfides (Figs. 3, 4). Tennantite-tetrahedrite has been identified at Nayongzhi but generally appears only in minor amounts near the feeder faults. Gangue minerals intergrown with the sulfides include hydrothermal dolomite, quartz, and minor calcite (Fig. 4).

Previous work on the deposit

The Nayongzhi deposit has been well studied since its discovery in 2015. Earlier works mostly focused on ore geology, ore formation, and the structural and stratigraphic controls on sulfide mineralization (Yang et al., 2018; Lan et al., 2019). Structural analysis indicates that deposit formation is associated with the NE- and NW-trending faults, directly linked to the Caledonian intracontinental orogeny. This association was confirmed by Rb-Sr dating of sphalerite, which yielded a Caledonian age (458 ± 2.9 Ma; Wang et al., 2018).

Fluid inclusion studies showed that the ore-forming fluids were of relatively low temperature (90° – 150° C) and moderate to high salinities (8.4–18.6 wt % NaCl equiv) and belonged to the H_2O -NaCl-MgCl- CO_2 system (Wei et al., 2024a). The observed fluid inclusion characteristics show evidence for mixing of a hotter ($>155^{\circ}$ C) and more saline (>18.0 wt % NaCl equiv) end member with a cooler ($<90^{\circ}$ C), low-salinity (<4.0 wt % NaCl equiv) end member. Spatially, homogenization temperatures decrease slightly ($\sim 35^{\circ}$ C) away from the feeder fault, with higher temperatures observed for the proximal discordant ores and lower temperatures in the distal stacked stratabound lenses (Wei et al., 2024a). The hotter fluid has been interpreted as a metal-rich basement-derived brine, which forms the ores on mixing with the lower-temperature, probably sulfur-rich, fluid.

Sulfur isotope studies indicate that reduced sulfur was sourced from the evaporitic sulfates present in the host Cambrian rocks (Jin et al., 2016; Zhou et al., 2018; Wei et al., 2024a). Finally, Pb and Cd isotope ratios of the sulfides essentially indicate the sourcing of the Pb and Zn from the Precambrian metamorphic basement (Song et al., 2024; Wei et al., 2024a). Lead isotope ratios also show a linear, systematic decrease with distance from the feeder fault, potentially indicating increased contributions from the host sediments through interactions with the ore fluids (Wei et al., 2024a).

Samples and Analytical Methods

Sample collection and preparation

Seventy ore samples were collected from the Nayongzhi deposit during three mapping campaigns conducted from 2015 to 2022. Most of these samples were collected in situ from different underground locations across all mining zones; some samples were collected from drill core intersections. A suite of 24 samples of ore and host rocks representative of all mining zones was selected (see App. Table A1 for details), with the precise sampling locations shown in Figure 1B. Nineteen of these 24 samples were prepared as polished 25-mm resin mounts or thin sections for petrography and microchemical investigations. Dolomite was sampled from 15 of these 24 samples, including host dolostones and ore-related dolomite (HD1), using a microdrill for carbon and oxygen isotope analysis. As indicated in Appendix Table A1, some of the samples used in this study had previously been used for fluid inclusion studies by Wei et al. (2024a).

Optical and cathodoluminescence (CL) microscopy

Petrographic observations were carried out at the State Key Laboratory of Ore Deposit Geochemistry, Institute of Geochemistry, Chinese Academy of Sciences (IGCAS). Optical and CL imaging were performed using a Reliotron Lumino-scope (HC1-LM) equipped with a polarization microscope and digital camera. The analyses were conducted at 5- to 10-kV acceleration voltage and a beam current of 0.5 mA under a vacuum of 80 to 120 mTorr.

Laser ablation-inductively coupled plasma-mass spectrometry (LA-ICP-MS)

Trace element contents of sphalerite were analyzed by LA-ICP-MS in two different laboratories in Australia and China (details in App. Table A2).

In Australia, a RESolution 193-nm ArF excimer laser ablation system coupled to an Agilent 7700x quadrupole mass spectrometer was used, housed at the Centre for Ore Deposit and Earth Sciences (CODES), University of Tasmania. Ablation was conducted in an atmosphere of pure He. Spot ablation was carried out using a 30- μm spot size at ~ 2.7 J/cm² with a 5 Hz pulse frequency. Typically, eight spot analyses were conducted on different mineral grains from each sample. For each analysis, 30 s of background signal was acquired, followed by 60 s of ablation. The following isotopes were monitored during each measurement: ²³Na, ²⁴Mg, ²⁷Al, ²⁹Si, ³⁴S, ⁴³Ca, ⁴⁹Ti, ⁵⁵Mn (dwell time 5 ms); ⁵¹V, ⁵³Cr, ⁹⁰Zr, ⁹³Nb, ⁹⁰Zr, ⁹³Nb, ¹⁵⁷Gd, ¹⁷⁸Hf, ¹⁸¹Ta, ²³²Th, ²³⁸U (dwell time 10 ms); and ⁵⁷Fe, ⁵⁹Co, ⁶⁰Ni, ⁶⁵Cu, ⁶⁶Zn, ⁷¹Ga, ⁷²Ge, ⁷⁵As, ⁷⁷Se, ¹⁰⁷Ag,

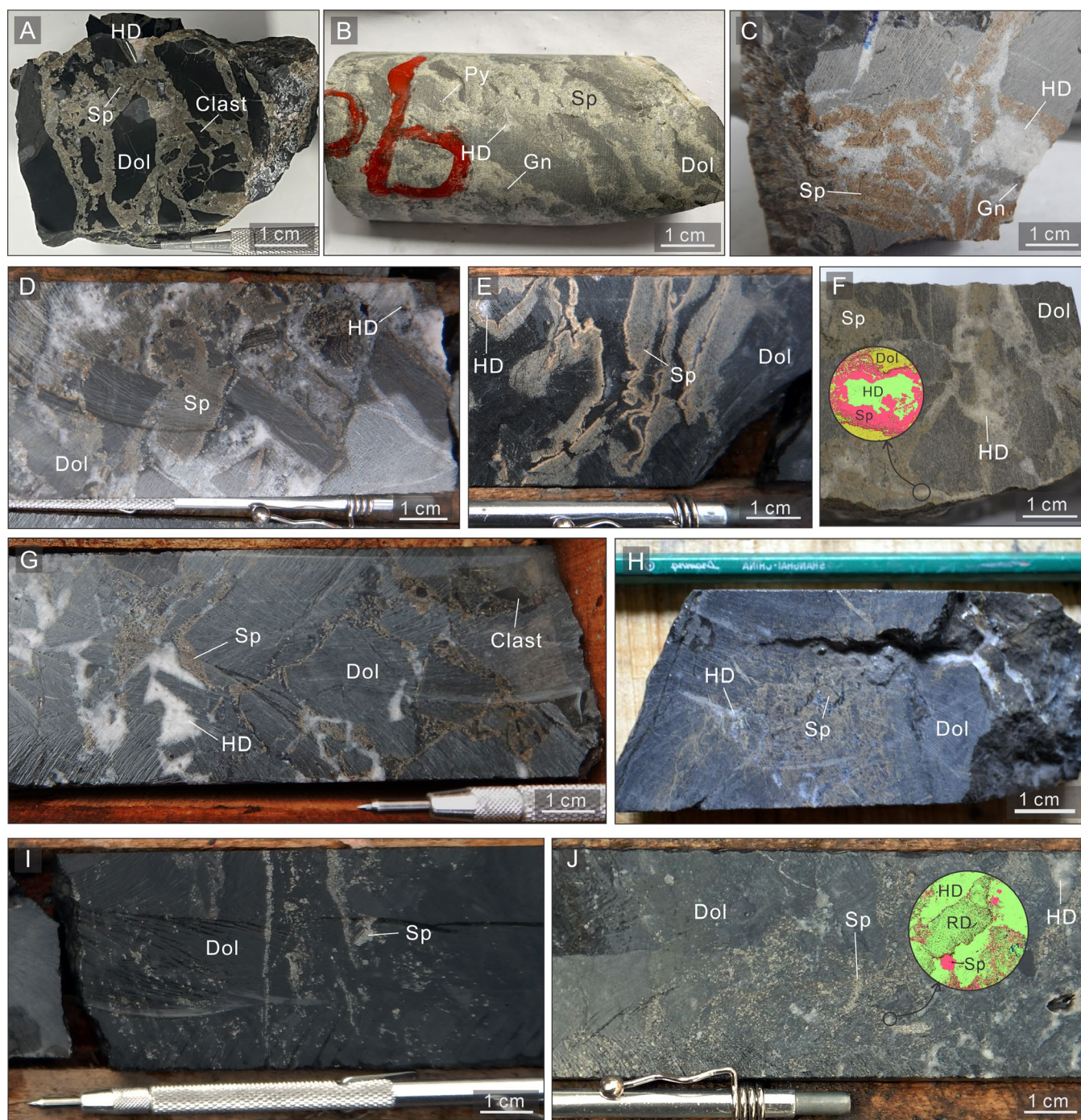


Fig. 3. Hand specimens and drill cores show the spatial variations in ore types, hydrothermal alteration, and zonation patterns. (A) Breccia ore consisting of large angular clasts cemented together by fine-grained sulfides (Yuhe, 1,220 m). (B) Sulfide ores cement the relic dolostone clasts (ZK3502, 557.2 m). (C) Sphalerite and hydrothermal dolomite infill open spaces within dolostone, forming interconnected networks and veinlets (Lumaolin, 1,270 m). (D) Sulfides infilling within the cavities of dissolution-collapse breccias and some clasts exhibit evidence of hydrothermal replacement (ZK2502, 207.8m). (E) Sulfide replacement of host dolostone locally obscures original sedimentary textures (ZK1402, 148.2 m). (F) Fractured dolostone was cemented by sulfides, with some clasts partially replaced. The inset highlights hydrothermal dolomite infilling cavities within sphalerite-dominated assemblages (Jinpo, 1,290m). (G) Sulfide and hydrothermal dolomite networks fill the host dolostone cracks (ZK002, 169.8m). (H) Host dolostone was selectively replaced by sphalerite-dominated sulfide assemblages (ZK3402, 131.2m). (I) Disseminated sphalerite within the dolostone (ZK3402, 228.4m). (J) Small pods of sulfides infill the preore dissolution porosity of dolostone breccias. The inset shows replacement dolomite surrounded by sphalerite pods and hydrothermal dolomite (ZK3401, 198.7m). Abbreviations: Dol = dolostone, Gn = galena, HD = hydrothermal dolomite, Py = pyrite, RD = replacement dolomite, Sp = sphalerite.

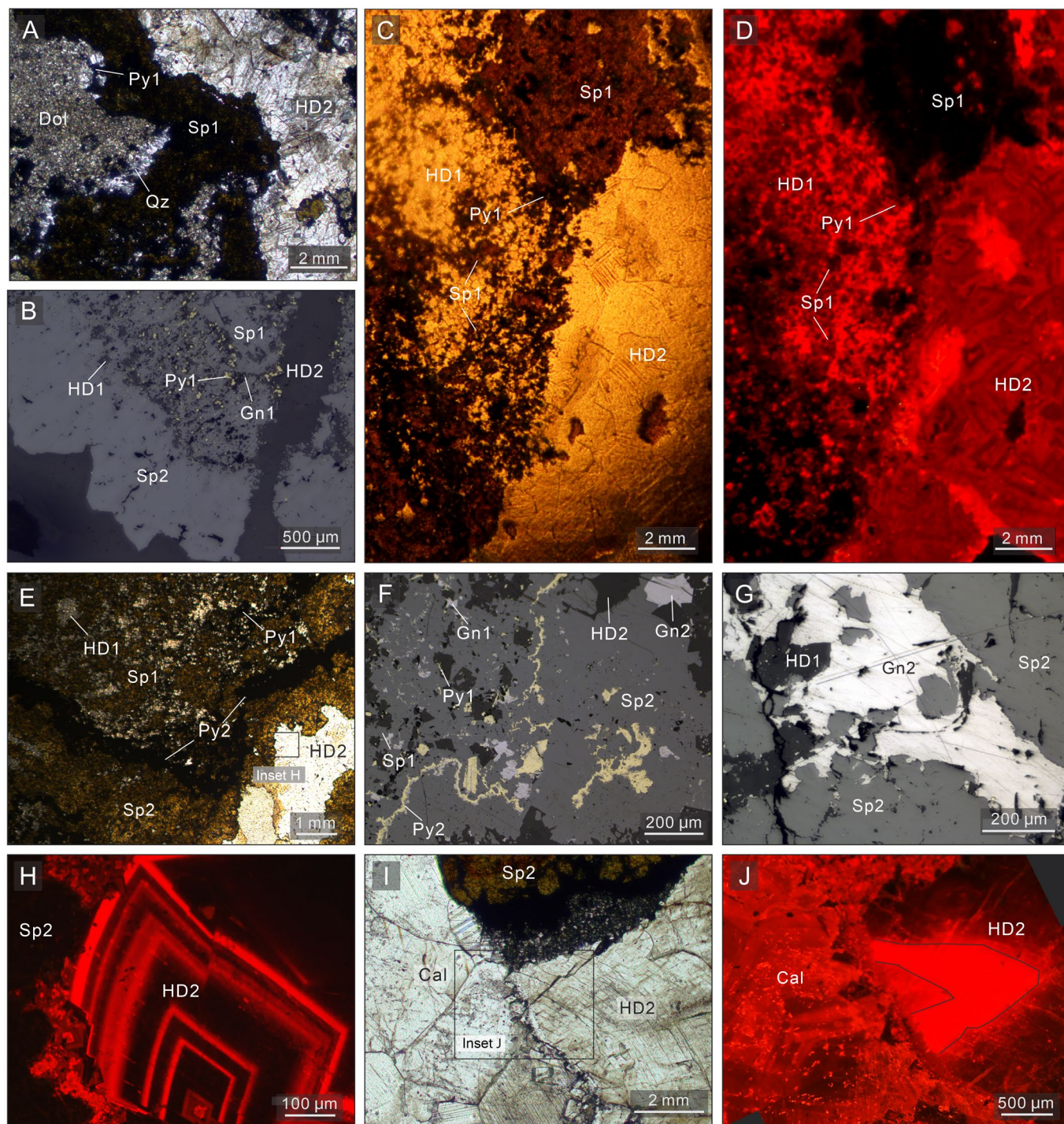


Fig. 4. Transmitted-light (A, C, E, and I), reflected-light (B, F, and G), and cathodoluminescence (CL) (D, H, and J) photomicrographs show textural variations of hydrothermal minerals from Nayongzhi. (A) Quartz and sulfides surround the edges of host rocks, and residual cavities are infilled with coarse-grained hydrothermal dolomite (HD2). (B) Fine-grained pyrite (Py1) is intergrown with porous sphalerite (Sp1), which is subsequently enclosed by coarse-crystalline sphalerite (Sp2). (C) Fine-grained hydrothermal dolomite (HD1) coexists with Py1 and Sp1, which is surrounded by saddle HD2 along the edges. (D) CL images highlight the different colors and textures between two generations of hydrothermal dolomite. (E) A thin layer of pyrite (Py2) surrounds the margin of porous Sp1, which is enclosed by Sp2. (F) High porous Sp1 is enclosed by harbor-like pyrite layers, which are intergrown with coarser, inclusion-free Sp2 and Gn2 along fractures. (G) Coarsely crystalline galena (Gn2) replaces Sp2, forming island-like galena remnants within the sphalerite. (H) Close-up CL images in (E) show individual HD2 crystals displaying growth zoning, characterized by alternating dark and light color bands. (I) Calcite vein crosscuts earlier Sp2 and saddle-shaped HD2. (J) CL images of Figure I show oscillatory color zoning in saddle-shaped HD2, distinct from the luminescence pattern of adjacent calcite. Abbreviations: Cal = calcite, Dol = dolostone, Gn = galena, HD = hydrothermal dolomite, Py = pyrite, Qz = quartz, Sp = sphalerite.

¹⁰⁹Ag, ¹¹¹Cd, ¹¹⁵In, ¹¹⁸Sn, ¹²¹Sb, ¹²⁵Te, ¹⁸²W, ¹⁹⁷Au, ²⁰²Hg, ²⁰⁵Tl, ²⁰⁶Pb, ²⁰⁷Pb, ²⁰⁸Pb, ²⁰⁹Bi (dwell time 20 ms). Reference materials STDGL2b2 (Danyushevsky et al., 2011) and GSD-1G (Jochum et al., 2008) were used for external calibration. A block with duplicate analyses of both reference materials was analyzed at the start of the session and after every 15 analyses of sample measurements.

In China, the analyses were performed on a RESolution 193-nm ArF excimer laser ablation system coupled to an Agilent 7700x ICP-MS instrument located at the State Key Laboratory of Ore Deposit Geochemistry, IGCAS. Ablation was conducted in an atmosphere of pure He using spot sizes of 26 μm with a laser repetition rate of 6 Hz. The laser beam fluence at the sample surface was $\sim 3.5 \text{ J/cm}^2$. Each analysis incorporated a background acquisition of approximately 20 s (gas blank) followed by 60 s of data acquisition and a final delay of 10 s for washout. The monitored isotopes were ²⁴Mg, ²⁷Al, ²⁹Si, ³⁴S, ⁴³Ca, ⁴⁵Sc, ⁴⁹Ti, ⁵¹V, ⁵³Cr, ⁵⁵Mn, ⁵⁷Fe, ⁶⁵Cu, ⁶⁶Zn, ⁷⁵As, ⁸⁵Rb, ²⁰⁸Pb (dwell time 5 ms); ⁸⁸Sr, ⁹⁵Mo (dwell time 6 ms); ⁷⁷Se, ¹⁰⁷Ag, ¹⁰⁹Ag, ¹¹¹Cd, ¹¹⁵In, ¹¹⁸Sn, ¹²¹Sb, ¹²⁵Te, ¹⁸⁴W, ²⁰⁵Tl, ²⁰¹Hg, ²⁰²Hg, ²⁰⁸Pb, ²⁰⁹Bi (dwell time 7 ms); and ⁵⁹Co, ⁶⁰Ni, ⁷¹Ga, ⁷²Ge, ⁷⁴Ge (dwell time 8 ms). Reference materials GSD-1G (Jochum et al., 2008) and UQAC-FeS (Savard et al., 2018) were used for external calibration. A block with duplicate analyses of both reference materials was analyzed at the start of the session and after every 15 to 20 analyses of sample measurements.

Data reduction was done using the LADR software package (Norris and Danyushevsky, 2018). Quantification was performed using ⁵⁷Fe as the internal standard element by normalizing content to achieve a total of 100 wt %. The contents of Cd and Zn were adjusted prior to normalization to account for their significant elemental fractionation relative to Fe (Danyushevsky et al., 2011). The contents of S and O are estimated based on monosulfide and oxide stoichiometry. The amount of oxygen in the sample varied between 700 and 3,000 ppm (App. Table A2), and thus the amounts of oxide components in the analyses are considered insignificant.

Isobaric interference of ¹¹⁵Sn on ¹¹⁵In and ⁷⁴Se on ⁷⁴Ge was corrected using known isotopic proportions of Sn and Se. Due to high contents of Ge in most samples, polyatomic interferences of ArS and FeO on Ge were considered insignificant; however, in a few analyses with Ge content <5 ppm their contents must be treated with caution. Interferences of ZrO and NbO on Ag were found to be insignificant due to low contents of Zr and Nb in the samples.

Large inclusions or mixed phases may affect the quality of LA-ICP-MS data. Micron-sized inclusions can be identified by signal anomalies with a width of several seconds and signals at least an order of magnitude higher than the background level (App. Fig. A1). In the samples from this study, a few sphalerite spots display signal anomalies of Fe, As, Se, Sn, Pb, Mg, Ca, and Mn, which are attributed to sulfide, sulfosalt, and carbonate inclusions (App. Fig. A1). The LADR software allows for the exclusion of spiky signals during quantitative integration, a standard practice to avoid contamination. Consequently, such data were excluded during data reduction to ensure accurate results from homogeneous signals (commonly >20 s). Full LA-ICP-MS results, detection limits, and uncertainties are detailed in Appendix Table A2.

Carbon and oxygen isotope analysis

The analysis of carbon and oxygen isotope compositions on the host dolostones and ore-related HD1 (detailed in App. Table A1) was performed on a Thermo Scientific MAT 253 mass spectrometer at the State Key Laboratory of Ore Deposit Geochemistry, IGCAS. Samples were quantitatively acidified in vacuo in an online autosampler using 100% orthophosphoric acid at 72°C for 24 h. After cryogenic purification of the generated carbon dioxide, the gas was transported to the mass spectrometer for analysis, with corrections applied using the phosphoric acid fractionation factor given by Rosenbaum and Sheppard (1986). All carbon and oxygen isotope ratios are expressed in per mil notation (‰) relative to the V-PDB and V-SMOW standards for carbon and oxygen, respectively. The internal accuracy of the analyses was $\pm 0.04\text{‰}$ (1 σ) for $\delta^{13}\text{C}$ and $\pm 0.03\text{‰}$ (1 σ) for $\delta^{18}\text{O}$, and the analytical precision for unknown samples was better than $\pm 0.05\text{‰}$ (1 σ) for $\delta^{13}\text{C}$ and $\pm 0.08\text{‰}$ (1 σ) for $\delta^{18}\text{O}$.

Data treatment and statistical analysis of trace element data sets

Statistical analysis was carried out to investigate the effects of sample locations, mineral associations, and generations on the trace element contents of sphalerite. The analysis mainly examined trace elements whose contents were above the detection limit in more than half of all individual spot analyses. Data pretreatment included (1) replacing the below detection limit values with the respective detection limit (Frenzel et al., 2016; Godefroy-Rodríguez et al., 2020); (2) calculating geometric mean values of each element in each generation of sphalerite per sample, following the guidelines for hierarchical data sets outlined by Frenzel (2023); and (3) log transformation of the geometric means to ensure data are approximately normally distributed (cf. Frenzel, 2023).

The data were then statistically analyzed using linear models (Winter, 2013). These models are designed to simultaneously account for the effects of different explanatory variables, allowing for the isolation of the most important controls (e.g., Dmitrijeva et al., 2018; Godefroy-Rodríguez et al., 2020). Using the notation employed to describe linear models in R (cf. Winter, 2013), the linear models we used were of the general form

$$X \sim \{Y_1, Y_2, \dots, Y_j\}, \quad (1)$$

where X is the dependent variable and Y_1 to Y_j are generic explanatory variables, either numeric or categorical. In the present case, the following specific combination of explanatory variables was considered:

$$\log(C_i) \sim \text{subsets of } \{\text{Distance, Level, RL, Generation, MA, Generation: Distance}\}. \quad (2)$$

Here, $\log(C_i)$ denotes the log-transformed element i contents. Distance, Level (i.e., elevation), and RL (relative level) are numeric variables, whereas Generation and MA (mineral association) are categorical. The factor *Generation: Distance* is an interaction term between the two variables Generation and Distance that allows for the fitting of generation-specific spatial gradients of the contents with distance.

To test the statistical significance of the different explanatory variables, a stepwise model fitting procedure was used

(Breaux, 1967). This method is widely used and straightforward for model selection (Desboulets, 2018). It involves sequentially removing insignificant variables (backward approach) or adding significant ones (forward approach). In this study, we used the forward approach, adding variables sequentially according to their explanatory power until no additional variables significantly improved the model at the $p < 0.05$ threshold. First, the explanatory variable with the highest R^2 in a simple model was chosen. Then, additional binary models were constructed by adding each remaining variable to the base model. Analysis of variance was used to compare these models to the base model to assess whether the explanatory performance improved significantly. The binary model with the lowest p value from this comparison was then chosen for the next iteration. This iterative process allowed us to estimate specific p values for the fixed effects (Winter, 2013). The R software suite version 4.3.3 (R Core Team, 2024) was used to conduct the analysis. An example script for one trace element is given in the Appendix Text.

Summary of Mineral Paragenesis

A generalized mineral paragenesis for the Nayongzhi deposit is shown in Figure 5. This summary is based on microscopic studies of different ore types for this study (Fig. 4), recent detailed observations by Wei et al. (2024a), and a review of other previous studies (Jin et al., 2016; Zhou et al., 2018).

Broadly, three ore stages can be recognized. The earliest sulfide assemblage is dominated by disseminated pyrite (Py1), porous sphalerite (Sp1), and fine-grained galena (Gn1) (Fig. 4A–D; Wei et al., 2024a). Py1 occurs as fine grains from 5 μm to several tens of micrometers, typically occupying interstitial spaces within porous Sp1 (Fig. 4A–F). It is characterized by relatively lower $\delta^{34}\text{S}$ values (3.6–10.5‰; Wei et al., 2024a) relative to Sp1 and Gn1 (19.7–27.9‰; Zhou et al., 2018; Wei et al., 2024a). Such values likely reflect a mixture of evaporite-derived sulfur and sulfur from ^{32}S -rich sedimentary/diagenetic pyrite (Wei et al. 2024a). The highly porous Sp1 is brown-yellow in transmitted light and varies in grain size from micrometers to millimeters (Fig. 4A–F). It commonly occurs as banded growths on the host dolostone (Fig. 4A) and as disseminations infilling intergranular porosity in dolomite (Fig. 4C, D). Sp1 encloses disseminated Py1 (Fig. 4B, F), indicating it precipitated slightly later than Py1. Fine-grained Gn1 is volumetrically minor and typically occurs as irregular, anhedral grains within the hydrothermal dolomite or enclosed by Sp1 (Fig. 4B, F). The sulfur isotopes of these sphalerite and galena resemble those derived from Cambrian evaporites through thermal sulfate reduction (19.7–27.9‰; Zhou et al., 2018; Wei et al., 2024a). Quartz is the earliest gangue mineral in the hydrothermal process, typically found at the interface between the host dolostone and porous Sp1 (Fig. 4A). Rhombs of hydrothermal dolomite (HD1) are overgrown and partially replaced by Sp1 (Fig. 4B, F). In most cases, spaces between HD1 grains are filled with sphalerite, and the HD1 shows slight recrystallization (Fig. 4C–E), exhibiting light-red, fine-grained shapes in the CL image (Fig. 4D).

The main ore stage predominantly contains coarsely crystalline sphalerite (Sp2) and galena (Gn2), with minor pyrite (Py2) (Fig. 4E–G; Wei et al., 2024a). These minerals fill the open spaces in the host dolostone (Fig. 4B, E–F) and replace

or overgrow the earlier sulfides (Fig. 4E, F). The earliest mineral at this stage is Py2, which typically occurs as a thin rim along the edges of porous Sp1 growths and is subsequently surrounded by coarsely crystalline Sp2 (Fig. 4E, F). Sp2 appears as light-yellow, inclusion-free crystals, typically several millimeters in diameter (Fig. 4B, E–G). It commonly overgrows the earlier Sp1 (Fig. 4B) or infills the fractures adjacent to Sp1, showing a distinct texture (Fig. 4F). In some cases, the thin layers of Py2 are replaced and corroded by later Sp2. The remnants of Py2 are preserved within the sphalerite, showing an embayment texture (Fig. 4F). Gn2 is subhedral to euhedral, medium to coarse grained, and typically replaces Sp2, indicating it precipitated later than sphalerite (Fig. 4F–G). Locally, tennantite-tetrahedrite is overgrown by later sphalerite and galena (Zhu, 2017). Sulfides in this stage have relatively heavier and more homogeneous $\delta^{34}\text{S}$ signatures relative to the earlier stage (15.2–25.5‰; Zhou et al., 2018; Wei et al., 2024a). The latest ore stage is marked by cavity- and vein-filling carbonate minerals, comprising both dolomite and calcite (Fig. 4E, H, I; Wei et al., 2024a). Hydrothermal dolomite (HD2) displays oscillatory zoning, characterized by alternating dark-red and light-red growth bands under CL images (Fig. 4D, H), distinguishing it from HD1. Minor calcite veins crosscut earlier hydrothermal minerals (Fig. 4I, J), indicating that calcite is the latest precipitate. Its homogeneous light-red CL response further distinguishes it from HD2 (Fig. 4J).

Mineral Chemistry and Stable Isotopes

Sphalerite trace element data

There are considerable variations in the contents of individual trace elements both within and between the analyzed sphalerite samples (App. Table A2). The results of LA-ICP-MS spot analyses of sphalerite, including 264 valid results from 19 samples, are summarized in Table 1. Specifically, the most commonly substituted trace elements within the sphalerite are Fe and Cd, with contents typically in the range from several thousand to tens of thousands of parts per million. Trace elements, including Cu, Ga, Ge, Sb, and Pb, are typically present in contents ranging from tens to hundreds of parts per million. Most of these elements are not unexpected, as they are interpreted to substitute for Zn in the sphalerite lattice, either directly or through coupled substitution (e.g., Cook et al., 2009; Ye et al., 2011). Trace elements such as Mn, Co, As, Ag, In, Sn, and Tl are present at contents around or below 10 ppm but are detected in more than two-thirds of the analyses. Nickel, Se, Te, and Bi were predominantly below detection limit (App. Table A2).

Dependence of sphalerite trace element contents on location and paragenetic stage

Figure 6 visualizes the relationships of individual trace elements with sphalerite generations and distance to the feeder faults. It is apparent that there are systematic compositional differences between Sp1 and Sp2 (Fig. 6; App. Fig. A2). The contents of Mn, Fe, Ge, Tl, and Pb are all higher in Sp1 (cf. Fig. 6), whereas Ga and Cd contents are higher in Sp2. Similar element distributions are also revealed in a linear transverse or within individual sphalerite grains, showing the cores to be enriched in Ge and Tl but depleted in Ga relative to

the rims (cf. App. Fig. A2). Furthermore, there appear to be systematic spatial variations in sphalerite composition across the deposit, with Mn, Fe, Cu, Ga, Ge, In, Tl, and possibly Pb contents decreasing with distance from the feeder fault, whereas Cd contents increase (Figs. 6, 7; App. Fig. A3).

This is corroborated by the results of our statistical analysis, summarized in Table 2, which provides p values for each effect (from sequential testing) and an R^2 value for the overall model for each trace element. Elements not shown in Table 2 were either mostly below detection limit (Ni, Se, Te, and Bi) or largely hosted by inclusions (e.g., As). The p values indicate that both distance and mineral generation have a statistically significant effect ($p < 2 \times 10^{-3}$) on the contents of most trace elements in sphalerite. However, exceptions arise with Cu and In, which are not significantly different between generations; Tl and Pb, which do not vary significantly with distance; and Co, Ag, Sn, and Sb, for which neither distance nor generation is significant. Overall, the fitted models account for between 28 and 83% of the total observed variances in trace element contents.

Sphalerite geothermometry and f_{S_2} calculation

Sphalerite crystallization temperatures were estimated through the GGIMFis (Ga, Ge, In, Mn, and Fe in sphalerite) geothermometer (Frenzel et al., 2016). The calculated results show Sp1 consistently formed at a relatively higher temperature (avg $95 \pm 71^\circ\text{--}150 \pm 50^\circ\text{C}$) compared with Sp2 (avg $80 \pm 78^\circ\text{--}115 \pm 63^\circ\text{C}$) across samples (Table 3). Although the estimates of absolute temperatures have relatively large uncertainties (Table 3), the differences between the two generations are well constrained, as shown by the PC1* values for the two generations, which do not overlap within samples. This is consistent with previous fluid inclusion studies of Nayongzhi sphalerite, showing Sp1 formed at 107° to 155°C and Sp2 formed at 95° to 122°C (Wei et al., 2024a). Importantly, the absolute temperatures and compositional uncertainties of sphalerite from GGIMFis for most samples (e.g., sample SY-10) closely matched the homogenization temperature of the corresponding fluid inclusions (Fig. 8A; Table 3), supporting the reliability of the GGIMFis estimates.

In addition to generations, the crystallization temperature of sphalerite varies significantly by sample location. The p values in Table 2 show a significant effect of the relative distance to feeder faults on temperatures ($p = 7.5 \times 10^{-5}$). As the distance to the feeder fault increases, the temperature generally decreases for both generations (Fig. 8A), albeit slightly more for Sp2.

Sulfur fugacity was estimated using the GGIMFis geothermometer and Fe contents in sphalerite (Frenzel et al., 2022), and the p values show statistical significance for the average difference between generations ($p < 2.0 \times 10^{-8}$; Table 2). Sp1 appears to have formed at a slightly higher sulfur fugacity (mean $\log_{10}f_{S_2}$ value between -23.9 ± 7.5 and -19.1 ± 4.1) compared with Sp2 (mean $\log_{10}f_{S_2}$ value between -24.9 ± 8.9 and -21.7 ± 6.1), though relatively large uncertainties are attached to the estimated absolute sulfur fugacity (Table 3). Spatially, the sulfur fugacity of sphalerite somewhat decreased with distance from the fluid flux center for both generations (Fig. 8B), but this trend is not statistically significant ($p = 0.02$; Table 2).

Table 1. Summary of LA-ICP-MS Spot Analyses of Trace Elements in Nayongzhi Sphalerite

Statistical parameter	Mn (ppm)	Fe (ppm)	Co (ppm)	Ni (ppm)	Cu (ppm)	Zn (wt %)	Ga (ppm)	Ge (ppm)	As (ppm)	Se (ppm)	Ag (ppm)	Cd (ppm)	In (ppm)	Sn (ppm)	Sb (ppm)	Te (ppm)	Tl (ppm)	Pb (ppm)	Bi (ppm)
$N = 264$	12.2	9,413	0.11	<0.16	20.0	65.4	37.2	401	1.52	<2.45	3.20	1,125	0.20	2.54	43.6	<0.57	14.5	516	<0.03
19 samples	Mean	9,67	8,852	0.15	<0.22	19.2	64.9	214	4.42	<2.17	5.45	1,102	0.18	2.15	27.9	<0.62	7.49	317	<0.03
	CI min	8,69	8,052	0.13	-	15.2	64.7	176	3.25	-	4.73	1,021	0.14	1.67	20.6	-	5.81	245	-
	CI max	10.8	9,732	0.18	-	24.3	65.2	260	6.03	-	6.28	1,188	0.24	2.77	37.7	-	9.65	411	-
	# BDL	8	0	89	231	7	4	2	74	254	0	0	55	26	11	208	14	3	246
	% of BDL	3	0	34	88	3	2	1	28	96	0	0	21	10	4	79	5	1	93
Detection limits for all	Min DL	0.21	8.19	0.02	0.10	0.17	0.74	0.02	0.34	1.03	0.02	0.12	0.00	0.03	0.05	0.24	0.01	0.05	0.01
	Max DL	0.97	32.0	0.09	0.33	0.74	2.37	0.08	0.84	3.54	0.24	0.44	0.03	0.23	0.38	1.44	0.10	0.22	0.08
LA-ICP-MS data	Median DL	0.46	13.0	0.04	0.14	0.35	1.17	0.06	0.47	2.45	0.11	0.30	0.01	0.10	0.19	0.47	0.04	0.08	0.03

Notes: Mean is geometric mean; CI min is the lower bound of the 95% confidence interval of the mean; CI max is the upper bound of the 95% confidence interval of the mean; # BDL is the number of analyzed values below the detection limit; % of BDL is the percentage of analyzed values below detection limit; Min DL is the minimum of detection limits; Max DL is the maximum of detection limits; Median DL is the median of detection limits; all statistics were calculated from measurements unaffected by inclusions (smooth ablation traces); values BDL were replaced by the DL for the calculations of the means and confidence intervals (cf. van den Boogaart and Tolosana-Delgado, 2013); the entire data set is listed in Appendix Table A2; dashes indicate no data available. Abbreviations: BDL = below detection limit, CI = confidence interval, DL = detection limit, LA-ICP-MS = laser ablation-inductively coupled plasma-mass spectrometry.

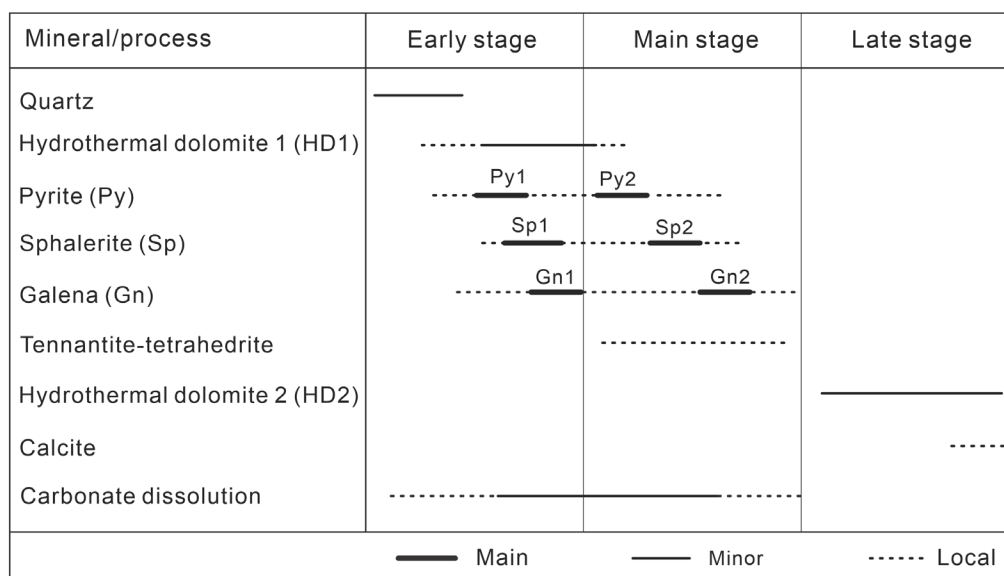


Fig. 5. Paragenetic sequence of the hydrothermal minerals in the Nayongzhi deposit.

Table 2. Summary of Linear Modeling Results for the Effects on Trace Elements in Nayongzhi Sphalerite

Variable	<i>p</i> Generations	<i>p</i> Distance	<i>p</i> Distance: Generations	<i>p</i> Mineral association	<i>p</i> RL	<i>p</i> Level	<i>R</i> ²
Co	0.4	1	0.75	0.73	0.63	0.72	0
Mn	8.4 × 10⁻⁶	3.8 × 10⁻¹²	0.43	0.72	0.64	0.25	0.78
Fe	2.6 × 10⁻⁹	5.8 × 10⁻¹³	0.51	0.3	0.66	0.71	0.83
Cu	0.23	4.3 × 10⁻⁶	0.55	0.08	0.94	0.03	0.44
Ga	9.5 × 10⁻⁹	6.2 × 10⁻⁸	0.46	0.56	0.27	0.62	0.76
Ge	2.2 × 10⁻⁷	5.1 × 10⁻⁷	0.02	0.59	0.91	0.55	0.68
Ag	0.46	0.23	0.85	0.96	3.1 × 10⁻⁴	5.6 × 10⁻⁸	0.44
Cd	1.3 × 10⁻¹⁰	7.1 × 10⁻⁸	0.17	0.93	0.76	0.46	0.77
In	0.73	6.7 × 10⁻⁹	0.64	0.74	0.12	4.0 × 10⁻³	0.68
Sn	0.49	0.19	0.67	0.07	0.1	0.64	0.16
Sb	0.32	0.41	0.73	0.92	0.15	0.67	0
Tl	8.0 × 10⁻⁴	1.0 × 10 ⁻³	0.98	0.24	0.34	0.54	0.4
Pb	4.0 × 10⁻³	0.01	0.91	0.36	0.57	0.52	0.28
T _{GGIMFis}	3.2 × 10⁻⁸	7.5 × 10⁻⁵	0.04	0.84	0.81	0.55	0.72
log ₁₀ (<i>f</i> _{S₂})	2.0 × 10⁻⁶	0.02	0.01	0.7	0.81	0.56	0.58
<i>df</i> _{S₂}	0.94	1.7 × 10⁻⁵	0.07	0.29	0.93	0.82	0.42

Notes: *p* values provide indications of statistical significance: the smaller the *p* value, the higher the probability of a relationship existing with the respective explanatory variable; bold values indicate significant relationships at a statistical significance level of $p < 2 \times 10^{-3}$; this significance level was adjusted downward from the standard level of $p < 0.05$ to account for multiple comparisons (cf. Dunn, 1961); *R*² denotes the coefficient of determination for the overall model (including all significant variables)

Abbreviations: GGIMFis = Ga, Ge, In, Mn, and Fe in sphalerite, RL = relative level

Carbon and oxygen isotope composition

The results of carbon and oxygen isotope compositions of carbonate samples, including two Cambrian host dolostones and 13 ore-related dolomites, are summarized together with previously reported data in Table 4 and presented graphically in Figure 9A. The $\delta^{13}\text{C}_{\text{V-PDB}}$ and $\delta^{18}\text{O}_{\text{V-SMOW}}$ values of the Cambrian marine dolostones we analyzed range from +0.6 to +0.8‰ and from +24.1 to +27.5‰, respectively ($n = 2$), in agreement with previous results ($\delta^{13}\text{C}_{\text{V-PDB}} = -1.7$ to +1.3‰ and $\delta^{18}\text{O}_{\text{V-SMOW}} = +22.9$ to +25.8‰ [$n = 8$]; Zhou et al., 2018). Ore-related HD1 has $\delta^{13}\text{C}_{\text{V-PDB}}$ values from -2.9 to -0.3‰ and $\delta^{18}\text{O}_{\text{V-SMOW}}$ values from +12.9 to +19.6‰ ($n = 13$). Al-

though the $\delta^{13}\text{C}_{\text{V-PDB}}$ values are comparable, the $\delta^{18}\text{O}_{\text{V-SMOW}}$ values are notably higher than those of previously reported ore-stage carbonate minerals ($\delta^{13}\text{C}_{\text{V-PDB}} = -4.1$ to +0.5‰ and $\delta^{18}\text{O}_{\text{V-SMOW}} = +12.4$ to +13.6‰ [$n = 7$]; Zhou et al., 2018).

The carbon and oxygen isotope compositions of ore-related dolomite vary systematically with distance from the feeder fault (Fig. 9A). Proximal to the feeder fault, $^{13}\text{C}_{\text{V-PDB}}$ and $^{18}\text{O}_{\text{V-SMOW}}$ are markedly depleted, whereas in distal regions, the isotopic compositions more closely resemble those of the host dolostones. The $\delta^{18}\text{O}$ values of hydrothermal fluids, calculated from the $\delta^{18}\text{O}_{\text{SMOW}}$ values of hydrothermal dolomite using fluid inclusion homogenization temperatures or the sphalerite

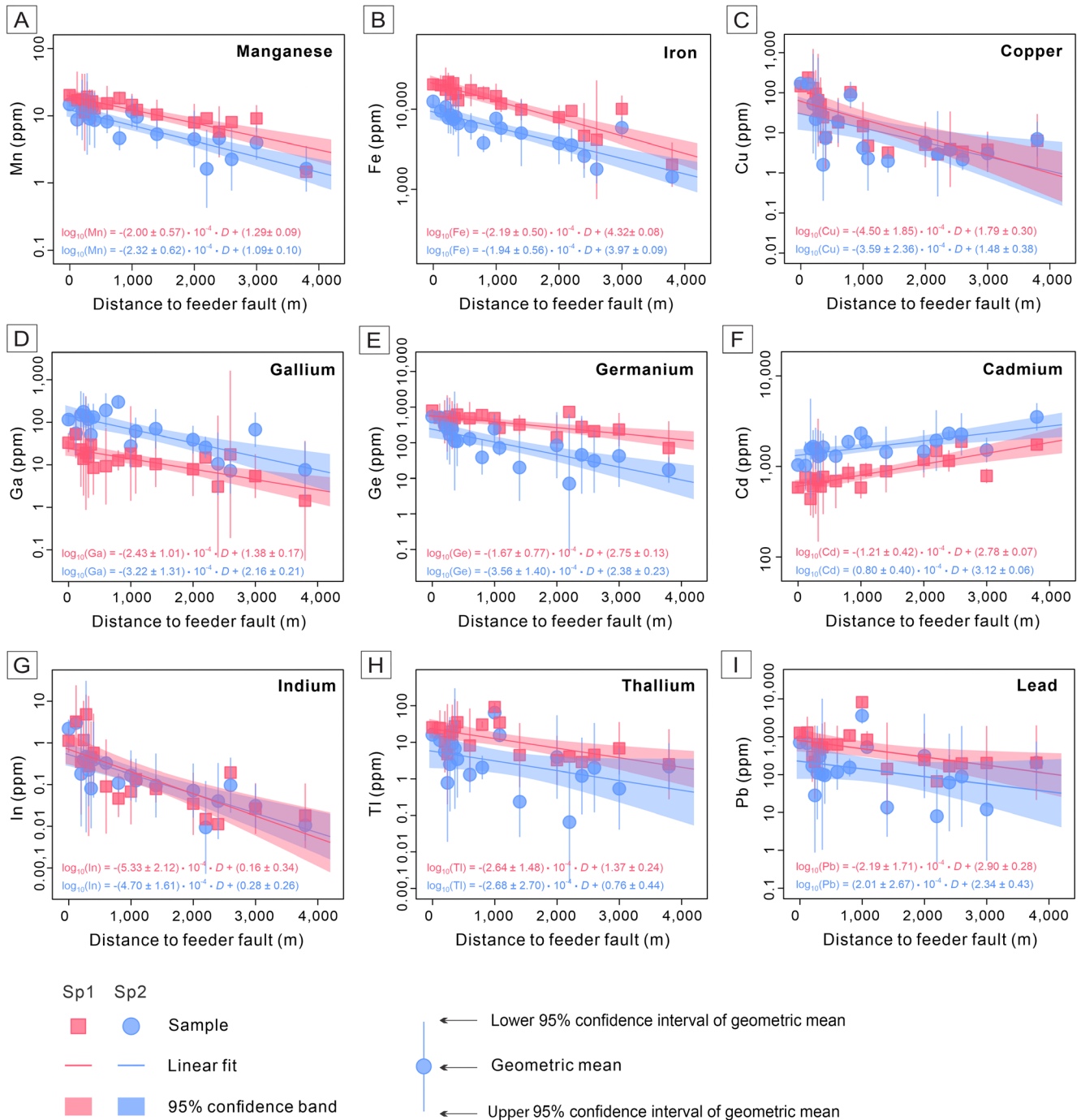


Fig. 6. Temporal and spatial variations of specific trace elements in Nayongzhi sphalerite, shown as geometric mean contents with corresponding 95% confidence intervals. (A) Mn. (B) Fe. (C) Cu. (D) Ga. (E) Ge. (F) Cd. (G) In. (H) Tl. (I) Pb. Cobalt, Ag, Sb, and Sn have been omitted from this figure since all explained effects were not statistically significant for these elements. Symbols represent the geometric mean, and solid lines indicate the 95% confidence intervals. The fit lines and corresponding 95% confidence bands are also shown in the plots.

GGIMF is geothermometer via the fractionation equation of Zheng (1999), range from -1.9 ± 0.9 to $+4.0 \pm 2.3\%$ (Table 4). When plotted on 2-D maps, these values exhibit spatial variations, becoming progressively heavier with increasing distance from the feeder fault (Fig. 9B).

Discussion

Sphalerite mineral chemistry and carbonate C-O isotope compositions show systematic variations across the Nayongzhi deposit, mirroring the observed mineralogical and metal zonation patterns reported in earlier work (Wei et al., 2024a). In

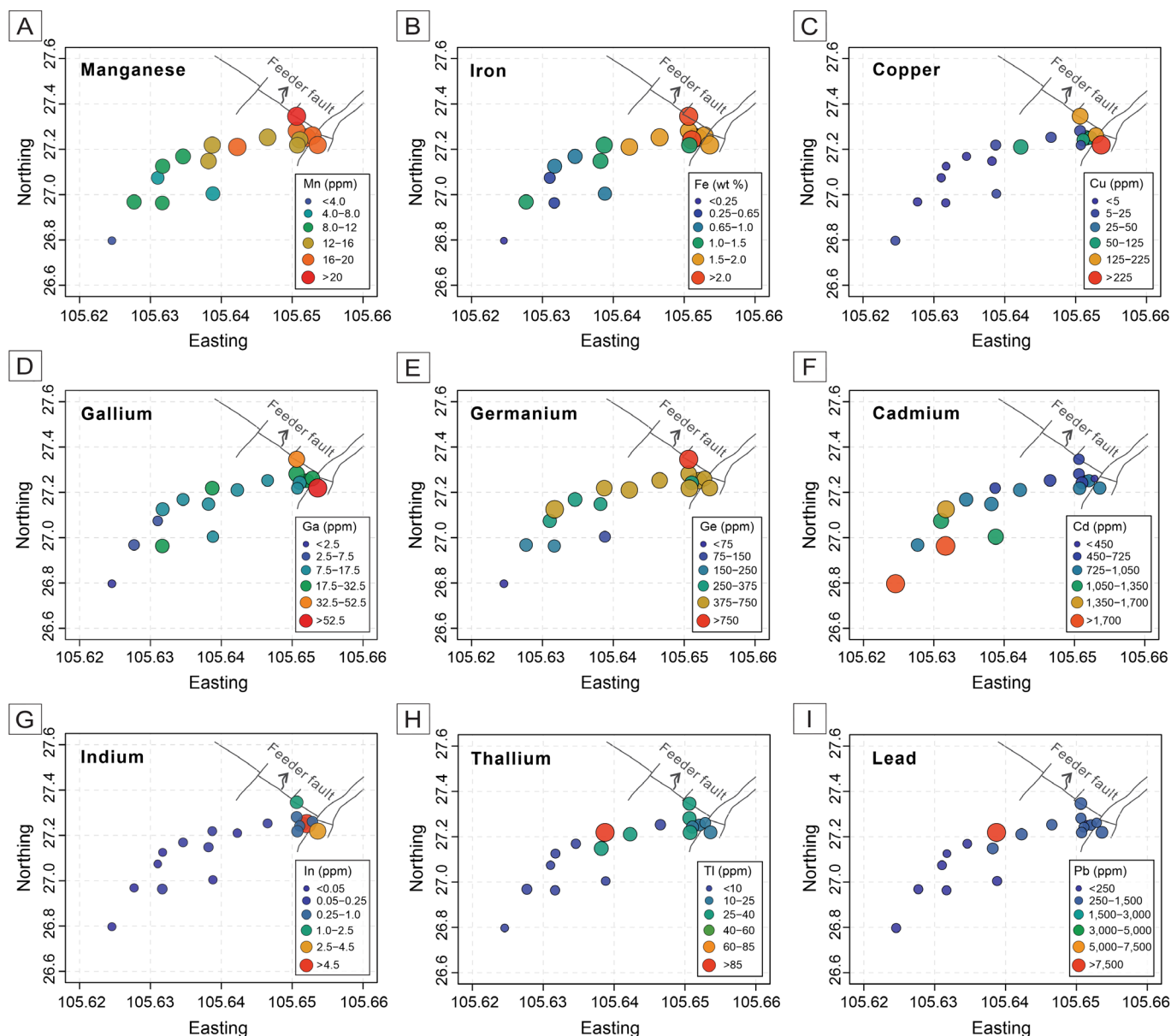


Fig. 7. Coordinate maps showing spatial variations of trace element contents in sphalerite samples from Nayongzhi. (A) Mn. (B) Fe. (C) Cu. (D) Ga. (E) Ge. (F) Cd. (G) In. (H) Tl. (I) Pb. The symbols represent the geometric means of early sphalerite. Gray solid lines show the locations of feeder structures of the Nayongzhi deposit.

the subsequent subsections, we examine the likely geologic controls on these new zonation patterns and then move on to the implications of the observed trends for tracing fluid flow paths at Nayongzhi and for mineral exploration in similar deposits across the world.

Control on carbonate C-O isotope zonation

The Nayongzhi hydrothermal dolomite samples from distal stratabound sulfide ores (e.g., samples SY-6 and SY-9) were formed through replacement of the host rocks and recrystallization and have heavier $\delta^{18}\text{O}_{\text{V-SMOW}}$ values, up to 19.6‰. These oxygen isotope values probably represent an inherited isotopic signature of the marine sedimentary carbonate rocks, as they are close to their isotopic compositions (~24.1–27.5‰; Fig.

9A; Table 4). Conversely, the hydrothermal dolomite proximal to the feeder fault has much lighter $\delta^{18}\text{O}_{\text{V-SMOW}}$ values of 14.1 to 14.2‰ (e.g., samples YH-15-4 and 1220-4) than the host rocks (Fig. 9A). These low $\delta^{18}\text{O}_{\text{V-SMOW}}$ values likely resulted from the influx of $\delta^{18}\text{O}$ -depleted fluids along feeder structure during dolomite crystallization, as evidenced by $\delta^{18}\text{O}_{\text{V-SMOW}}$ values as low as $-1.9 \pm 0.9\text{‰}$ (Table 4) in sample YH-15-2 directly found in the feeder fault. This value is likely closer to the composition of the deep fluid influx, possibly derived from original Ordovician seawater ($\delta^{18}\text{O} \leq -2\text{‰}$; Veizer et al., 1999; Bindeman, 2021). Overall, the observed spatial $\delta^{18}\text{O}$ variations in the fluids appear to be governed by oxygen isotope exchange between a deeper $\delta^{18}\text{O}$ -depleted fluid and the $\delta^{18}\text{O}$ -enriched host rocks. Specifically, the influx of the deep

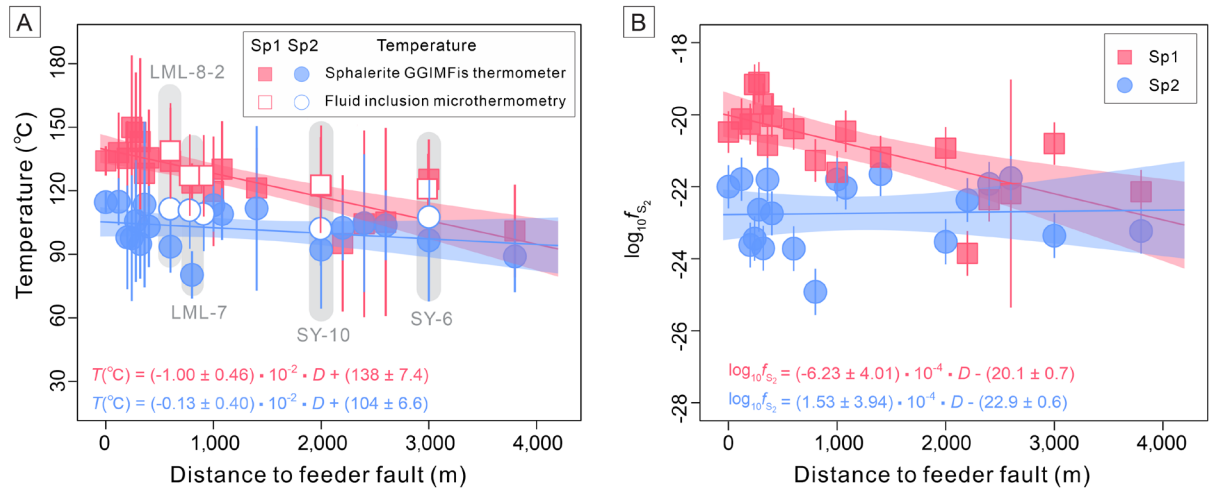


Fig. 8. Temporal and spatial variations of (A) crystallization temperature and (B) $\log_{10}(f_{S_2})$ in Nayongzhi. The homogenization temperature of fluid inclusions is also included in (A) for comparison (data from Wei et al., 2024a). Abbreviation: GGIMFis = Ga, Ge, In, Mn, and Fe in sphalerite.

Table 3. Crystallization Temperature and $\log_{10}(f_{S_2})$ Values Estimated From Nayongzhi Sphalerite Chemistry Compositions (arithmetic means $\pm 2\sigma$)

Sample ID	Easting	Northing	Elevation (m)	Distance to feeder fault (m)	Generation	Number of analyses	PC1*	T_{GGIMFis} (°C)	$2\sigma_{\text{external}}$ (°C)	$\log_{10}(\text{FeS})$ (mol %)	$\log_{10}(f_{S_2})$	$2\sigma_{\text{external}}(f_{S_2})$	Temperature for fluid inclusion (°C)
1220-1	105°39'5.62"	27°15'4.64"	1,220	241	Sp1	4	1.07 $\pm$ 0.63	150 $\pm$ 34	$\pm$ 50	-0.88 $\pm$ 0.05	-19.2 $\pm$ 0.56	$\pm$ 4.0	
					Sp2	5	2.04 $\pm$ 0.55	98 $\pm$ 30	$\pm$ 70	-1.24 $\pm$ 0.11	-23.4 $\pm$ 0.62	$\pm$ 7.3	
1220-2	105°39'7.42"	27°15'5.72"	1,220	280	Sp1	4	1.11 $\pm$ 0.52	148 $\pm$ 28	$\pm$ 50	-1.01 $\pm$ 0.11	-19.1 $\pm$ 0.56	$\pm$ 4.1	
					Sp2	4	1.89 $\pm$ 0.53	106 $\pm$ 29	$\pm$ 67	-1.21 $\pm$ 0.12	-22.6 $\pm$ 0.61	$\pm$ 6.6	
1220-3	105°39'2.27"	27°16'52.8"	1,220	360	Sp1	7	1.47 $\pm$ 0.41	128 $\pm$ 22	$\pm$ 57	-1.00 $\pm$ 0.12	-20.8 $\pm$ 0.58	$\pm$ 5.1	
					Sp2	5	1.74 $\pm$ 0.72	114 $\pm$ 39	$\pm$ 63	-1.23 $\pm$ 0.11	-21.8 $\pm$ 0.60	$\pm$ 6.1	
1220-4	105°39'10.3"	27°15'40.2"	1,220	200	Sp1	4	1.34 $\pm$ 0.32	135 $\pm$ 18	$\pm$ 55	-0.96 $\pm$ 0.14	-20.3 $\pm$ 0.57	$\pm$ 4.7	
					Sp2	4	2.03 $\pm$ 0.45	98 $\pm$ 25	$\pm$ 70	-1.13 $\pm$ 0.09	-23.6 $\pm$ 0.62	$\pm$ 7.3	
1390-11	105°38'4.56"	27°10'7.21"	1,390	1,401	Sp1	11	1.60 $\pm$ 0.35	121 $\pm$ 19	$\pm$ 60	-1.16 $\pm$ 0.10	-21.2 $\pm$ 0.59	$\pm$ 5.5	
					Sp2	6	1.78 $\pm$ 0.72	112 $\pm$ 39	$\pm$ 64	-1.39 $\pm$ 0.25	-21.7 $\pm$ 0.61	$\pm$ 6.1	
1390-2	105°38'17.5"	27°8'52.8"	1,390	1,080	Sp1	11	1.44 $\pm$ 0.42	130 $\pm$ 23	$\pm$ 57	-1.10 $\pm$ 0.07	-20.5 $\pm$ 0.58	$\pm$ 5.0	
					Sp2	8	1.83 $\pm$ 0.23	109 $\pm$ 12	$\pm$ 65	-1.34 $\pm$ 0.12	-22.0 $\pm$ 0.60	$\pm$ 6.4	
JP-1	105°38'19.4"	27°13'7.21"	1,390	1,000	Sp1	5	1.63 $\pm$ 0.48	120 $\pm$ 26	$\pm$ 61	-1.02 $\pm$ 0.09	-21.6 $\pm$ 0.59	$\pm$ 5.7	
					Sp2	7	1.75 $\pm$ 0.23	113 $\pm$ 12	$\pm$ 64	-1.25 $\pm$ 0.04	-21.8 $\pm$ 0.60	$\pm$ 6.1	
JP-21	105°37'51.7"	27°4'26.4"	1,390	2,600	Sp1	4	1.91 $\pm$ 0.81	104 $\pm$ 44	$\pm$ 67	-1.42 $\pm$ 0.20	-22.4 $\pm$ 0.61	$\pm$ 6.8	
					Sp2	6	1.91 $\pm$ 0.60	105 $\pm$ 33	$\pm$ 67	-1.62 $\pm$ 0.17	-21.9 $\pm$ 0.61	$\pm$ 6.8	
JP-5	105°37'54.2"	27°7'31.2"	1,290	2,200	Sp1	2	2.08 $\pm$ 0.59	95 $\pm$ 32	$\pm$ 71	-1.17 $\pm$ 0.40	-23.9 $\pm$ 0.62	$\pm$ 7.5	
					Sp2	4	1.95 $\pm$ 0.29	103 $\pm$ 15	$\pm$ 68	-1.51 $\pm$ 0.10	-22.4 $\pm$ 0.61	$\pm$ 6.0	
LML-7 ¹	105°38'32.1"	27°12'37.2"	1,270	800	Sp1	21	1.56 $\pm$ 0.17	124 $\pm$ 9	$\pm$ 59	-1.00 $\pm$ 0.07	-21.3 $\pm$ 0.59	$\pm$ 5.4	128 $\pm$ 12
					Sp2	8	2.36 $\pm$ 0.21	80 $\pm$ 11	$\pm$ 78	-1.49 $\pm$ 0.08	-24.9 $\pm$ 0.64	$\pm$ 8.9	111 $\pm$ 10
LML-8-2 ¹	105°38'47.5"	27°15'9.6"	1,270	601	Sp1	6	1.37 $\pm$ 0.46	134 $\pm$ 25	$\pm$ 55	-0.97 $\pm$ 0.12	-20.4 $\pm$ 0.58	$\pm$ 4.8	139 $\pm$ 13
					Sp2	6	2.11 $\pm$ 0.23	94 $\pm$ 12	$\pm$ 72	-1.32 $\pm$ 0.10	-23.7 $\pm$ 0.62	$\pm$ 7.7	111 $\pm$ 9.5
SY-10 ¹	105°38'19.7"	27°0'14.4"	1,270	2,000	Sp1	7	1.58 $\pm$ 0.53	122 $\pm$ 28	$\pm$ 60	-1.23 $\pm$ 0.10	-20.9 $\pm$ 0.59	$\pm$ 5.5	123 $\pm$ 14
					Sp2	7	2.14 $\pm$ 0.51	92 $\pm$ 28	$\pm$ 73	-1.49 $\pm$ 0.17	-23.5 $\pm$ 0.63	$\pm$ 7.8	106 $\pm$ 11
SY-6 ¹	105°37'39.7"	26°58'4.80"	1,290	3,000	Sp1	9	1.52 $\pm$ 0.35	125 $\pm$ 19	$\pm$ 59	-1.15 $\pm$ 0.11	-20.8 $\pm$ 0.59	$\pm$ 5.3	116 $\pm$ 11
					Sp2	7	2.06 $\pm$ 0.53	97 $\pm$ 29	$\pm$ 71	-1.34 $\pm$ 0.09	-23.4 $\pm$ 0.62	$\pm$ 7.4	109 $\pm$ 11
SY-9	105°37'28.4"	26°47'48.0"	1,390	3,799	Sp1	4	1.97 $\pm$ 0.40	101 $\pm$ 22	$\pm$ 69	-1.71 $\pm$ 0.14	-22.1 $\pm$ 0.61	$\pm$ 6.7	
					Sp2	5	2.20 $\pm$ 0.31	89 $\pm$ 17	$\pm$ 74	-1.83 $\pm$ 0.05	-23.2 $\pm$ 0.63	$\pm$ 8.1	
YH1175-2	105°39'3.82"	27°14'30.0"	1,175	320	Sp1	4	1.20 $\pm$ 0.73	143 $\pm$ 39	$\pm$ 52	-0.90 $\pm$ 0.04	-19.7 $\pm$ 0.57	$\pm$ 4.3	
					Sp2	6	2.09 $\pm$ 0.30	95 $\pm$ 16	$\pm$ 72	-1.25 $\pm$ 0.12	-23.7 $\pm$ 0.62	$\pm$ 7.6	
YH-15-2	105°39'2.27"	27°20'45.6"	1,220	0	Sp1	17	1.36 $\pm$ 0.13	134 $\pm$ 7	$\pm$ 55	-0.91 $\pm$ 0.06	-20.5 $\pm$ 0.58	$\pm$ 4.8	
					Sp2	20	1.72 $\pm$ 0.09	115 $\pm$ 5	$\pm$ 63	-1.08 $\pm$ 0.06	-22.0 $\pm$ 0.60	$\pm$ 6.0	
YH-15-4	105°39'2.63"	27°13'4.80"	1,175	400	Sp1	6	1.34 $\pm$ 0.42	136 $\pm$ 23	$\pm$ 55	-1.07 $\pm$ 0.13	-20.0 $\pm$ 0.58	$\pm$ 4.7	
					Sp2	6	1.93 $\pm$ 0.35	103 $\pm$ 19	$\pm$ 68	-1.30 $\pm$ 0.25	-22.7 $\pm$ 0.61	$\pm$ 6.9	
ZK202-1	105°37'54.0"	26°57'48.0"	1,280	2,400	Sp1	3	1.90 $\pm$ 0.82	105 $\pm$ 44	$\pm$ 67	-1.46 $\pm$ 0.09	-22.2 $\pm$ 0.61	$\pm$ 6.9	
					Sp2	5	1.92 $\pm$ 0.30	104 $\pm$ 16	$\pm$ 67	-1.75 $\pm$ 0.11	-21.8 $\pm$ 0.61	$\pm$ 6.9	
ZK3502	105°39'13.0"	27°13'8.40"	818	120	Sp1	7	1.29 $\pm$ 0.35	138 $\pm$ 19	$\pm$ 54	-0.92 $\pm$ 0.09	-20.1 $\pm$ 0.57	$\pm$ 4.6	

Notes: The uncertainties for PC1*, T_{GGIMFis} , $\log_{10}(\text{FeS})$, and $\log_{10}(f_{S_2})$ in the respective columns are internal uncertainties only, resulting from compositional variations in sphalerite within the generations of each sample; external uncertainties shown as $2\sigma_{\text{external}}$ in separate columns contain all sources of error, including the calibration of the GGIMFis geothermometer (Frenzel et al., 2016) and the model for $\log_{10}(f_{S_2})$ based on sphalerite composition (cf. Frenzel et al., 2022); the uncertainty levels correspond to ± 2 standard errors; for $\log_{10}(f_{S_2})$, most external uncertainty arises from the uncertainty in absolute GGIMFis temperature.

Abbreviations: GGIMFis = Ga, Ge, In, Mn, and Fe in sphalerite; Sp = sphalerite

¹Samples provided include homogenization temperatures of fluid inclusions in sphalerite from Wei et al. (2024a) for comparison with corresponding GGIMFis temperature

fluid along the feeder fault results in higher fluid-rock ratios at and near the fault and lower fluid-rock ratios farther away (Fig. 9B). These ratios are reflected in the isotopic compositions of the hydrothermal dolomite.

The C isotope compositions only exhibit significant variations away from the host-rock signature in and closer to the feeder fault, where observed $\delta^{13}\text{C}_{\text{V-PDB}}$ values are relatively lighter (up to -2.9‰ ; Table 4, Fig. 9A). The origin of the

light carbon is not clear. It may reflect the compositions of Ordovician seawater ($\delta^{13}\text{C}_{\text{V-PDB}} < 0\text{‰}$; Veizer et al., 1999) or be derived from the rocks underlying the host rocks ($\delta^{13}\text{C}_{\text{V-PDB}}$ down to -10‰ ; Shields and Veizer, 2002). In addition, the contribution of organic C cannot be ruled out.

Compared with oxygen isotopes, $\delta^{13}\text{C}_{\text{V-PDB}}$ variations ($< 2.0\text{‰}$) show a much smaller footprint. This discrepancy may stem from the low abundance of C-bearing species in the deep fluid (e.g., $< 0.5 \text{ mol/L CO}_2$; Duan et al., 2006) relative to the host rocks. During fluid-rock interaction, the host-rock signatures still dominate for carbon, even at high fluid-rock ratios. In contrast, oxygen, mainly as H_2O , is abundant in the deep fluid influx, which may more effectively change the O isotope compositions of the host rocks (Vazquez et al., 1998). As a consequence, oxygen isotope changes remain discernible at lower fluid-rock ratios in the distal parts of the deposit (Fig. 9B).

In summary, both C and O isotopes are laterally zoned away from the feeder fault but $\delta^{18}\text{O}$ shows a much larger footprint than $\delta^{13}\text{C}$. Similar O isotope zonation has been observed in other carbonate-hosted ore deposit types (e.g., Naito et al., 1995; Barker et al., 2013). The $\delta^{18}\text{O}$ values of hydrothermal dolomite are expected to be an effective tracer for vectoring fluid flow paths.

Temporal and spatial variations in sphalerite composition

Trace elements in sphalerite display notable spatiotemporal variations (Figs. 6, 7; App. Fig. A3). These variations may be attributable to a variety of factors, including changes in temperature, f_{S_2} , pH, f_{O_2} , fluid compositions, or the combined effect of several of these factors, driven by fluid mixing or fluid-rock interactions. In the following subsections, we quantitatively evaluate the observable correlations of trace element contents with temperature and df_{S_2} (corrected value for $\log_{10}(f_{\text{S}_2})$ after decorrelation with $1/T$) and examine their probable effects on the observed spatiotemporal patterns. Other potential controls will also be considered. We start with the observed temporal variations and their controls and then move on to address the likely controls behind the observed spatial variations.

Temporal variations: As noted earlier, early formed Sp1 is enriched in Mn, Fe, Ge, Tl, and Pb but depleted in Ga and Cd relative to late Sp2 (Fig. 6; App. Fig. A2). The two generations also differ in formation temperatures and estimated sulfur fugacities (f_{S_2}) (Fig. 8; Table 3). This raises the obvious question to what extent the observed differences may be explained by differences in formation temperature and f_{S_2} .

Sulfur fugacity commonly varies strongly with temperature, even when the sulfidation state of a fluid remains constant (Barton and Toulmin, 1966; Scott and Barnes, 1971; Scott and Kissin, 1973). In the present data set, this is evidenced by the strong correlation between temperature and $\log_{10}(f_{\text{S}_2})$ ($R^2 = 0.94$, $p < 0.001$; Fig. 10). Without an appropriate correction, it would therefore be impossible to distinguish between the specific effects of temperature and f_{S_2} . The corrections of the temperature effect were done by fitting a model of the following form to the data set (cf. Frenzel et al., 2022):

$$\log_{10}(f_{\text{S}_2}) = \frac{A}{T} + B. \quad (3)$$

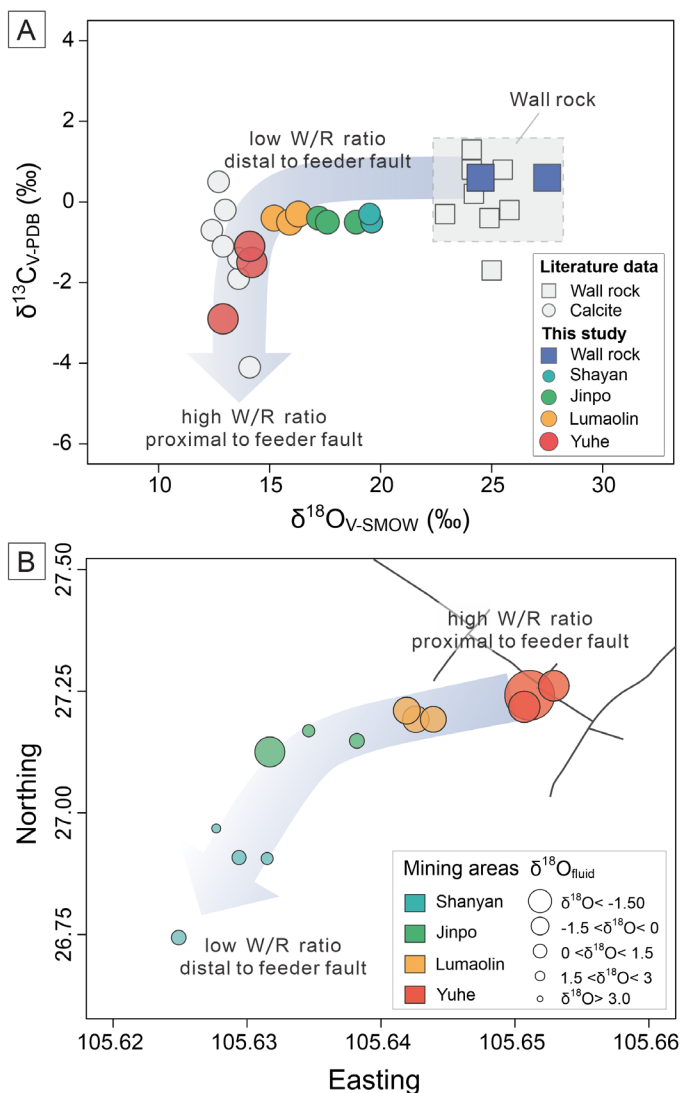


Fig. 9. Spatial variations of carbon and oxygen isotopes in carbonates from Nayongzhi. (A) Plot of carbon versus oxygen isotope for ore-related carbonate minerals and the host dolostone. Gray symbols indicate data from Zhou et al. (2018). (B) Coordinate map showing spatial variations in calculated oxygen isotope composition relative to the distance from the feeder structure. Abbreviations: V-PDB = Vienna-Pee Dee Belemnite, V-SMOW = Vienna-standard mean ocean water, W/R = water-rock.

The fitted model and its coefficients A and B are shown in Figure 10A. The residuals of this model correspond to the corrected df_{S_2} values shown in Figure 10B and were used for further data analysis.

The p value ($p = 0.94$) in Table 2 indicates that df_{S_2} values are not significantly different between the two generations of sphalerite. The plotted trace element contents for different samples across generations against df_{S_2} (Fig. 11) further show that df_{S_2} cannot account for the differences in trace element contents between sphalerite generations because of the overlapping df_{S_2} values. In contrast, the content differences of those trace elements that show marked differences between the two generations (i.e., Mn, Fe, Ga, Ge, Cd, In, Tl, and possibly Pb) could be related to temperature changes instead (Fig. 12).

Table 4. Carbon and Oxygen Isotope Compositions of Carbonate Minerals and Host Dolostone From Nayongzhi

Sample ID	Easting	Northing	Elevation (m)	Distance to feeder fault (m)	Mineral	$\delta^{13}\text{C}_{\text{PDB}}$ (‰)	$\delta^{18}\text{O}_{\text{SMOW}}$ (‰)	$\delta^{18}\text{O}_{\text{PDB}}$ (‰)	$\delta^{18}\text{O}_{\text{fluid}}$ (‰)	Temperature (°C)
JP-22	105°37'51.7"	27°4'26.4"	1,390	2,800	Dolostone	0.8	24.1	-6.6		
SY-21	105°37'45.7"	26°54'30.0"	1,370	2,770	Dolostone	0.6	27.5	-3.3		
TW-35-1 ¹					Dolostone	-0.2	25.8	-4.9		
C-11					Dolostone	-0.3	22.9	-7.7		
C-21					Dolostone	-0.4	24.9	-5.8		
YH1220-1 ¹					Dolostone	0.2	24.2	-6.5		
YH1220-2 ¹					Dolostone	0.6	24.5	-6.2		
YH1220-3 ¹					Dolostone	0.8	25.5	-5.2		
YH1220-4 ¹					Dolostone	1.3	24.1	-6.6		
YH1220-5 ¹					Dolostone	-1.7	25	-5.7		
YH-15-2	105°39'3.82"	27°14'30.0"	1,220	0	HD1	-2.9	12.9	-17.4	-1.9 ± 0.9	134 ± 7
1220-4	105°39'10.3"	27°15'40.2"	1,220	200	HD1	-1.5	14.2	-16.2	-0.5 ± 2.2	135 ± 18
YH-15-4	105°39'2.63"	27°13'4.8"	1,175	400	HD1	-1.1	14.1	-16.3	-0.5 ± 2.8	136 ± 23
LML-8 ²	105°38'33.3"	27°11'32.8"	1,270	780	HD1	-0.4	15.2	-15.2	0.9 ± 1.6	139 ± 13
LML-7 ²	105°38'32.1"	27°12'37.2"	1,270	800	HD1	-0.5	15.9	-14.5	0.8 ± 1.1	131 ± 9
LML-9 ²	105°38'37.5"	27°11'32.4"	1,270	820	HD1	-0.3	16.3	-14.1	1.0 ± 1.6	129 ± 13
1390-2	105°38'17.5"	27°8'52.8"	1,390	1,080	HD1	-0.4	17.2	-13.3	2.0 ± 2.8	130 ± 23
1390-11	105°38'4.56"	27°10'7.21"	1,390	1,401	HD1	-0.5	18.9	-11.6	2.9 ± 2.3	121 ± 19
JP-5	105°37'54.2"	27°7'31.2"	1,290	2,200	HD1	-0.5	17.6	-12.9	-1.4 ± 3.8	95 ± 32
SY-21 ²	105°37'45.7"	26°54'30.0"	1,370	2,770	HD1	-0.5	19.6	-10.9	3.0 ± 1.2	116 ± 10
SY-33 ²	105°37'53.3"	26°54'22.8"	1,390	2,880	HD1	-0.5	19.6	-10.9	3.7 ± 0.7	122 ± 6
SY-6	105°37'39.7"	26°58'4.80"	1,290	3,000	HD1	-0.5	19.6	-10.9	4.0 ± 2.3	125 ± 19
SY-9	105°37'28.4"	26°47'48.0"	1,390	3,799	HD1	-0.3	19.5	-11.0	1.2 ± 2.4	101 ± 20
H06-61					Calcite	-4.1	14.1	-16.3		
H06-17 ¹					Calcite	-0.7	12.4	-17.9		
1290F-4 ¹					Calcite	-1.9	13.6	-16.7		
1390Cal ¹					Calcite	-0.2	13	-17.3		
1390JP-4 ¹					Calcite	-1.4	13.6	-16.7		
1390JP-6 ¹					Calcite	-1.1	12.9	-17.4		
1390JP-7 ¹					Calcite	0.5	12.7	-17.6		

Notes: For samples analyzed in this study, oxygen isotopes in hydrothermal fluids were calculated using the fractionation equation of Zheng (1999); temperatures were determined using the GGIMFis geothermometer for coexisting sphalerite (Frenzel et al., 2016)

Abbreviations: GGIMFis = Ga, Ge, In, Mn, and Fe in sphalerite, HD = hydrothermal dolomite, PDB = Pee Dee Belemnite, SMOW = standard mean ocean water

¹Samples indicate data from Zhou et al. (2018), mostly carbon-oxygen isotopes in the host dolostones and calcite

²Fluid inclusion data from carbonate minerals and co-existing sphalerite were used

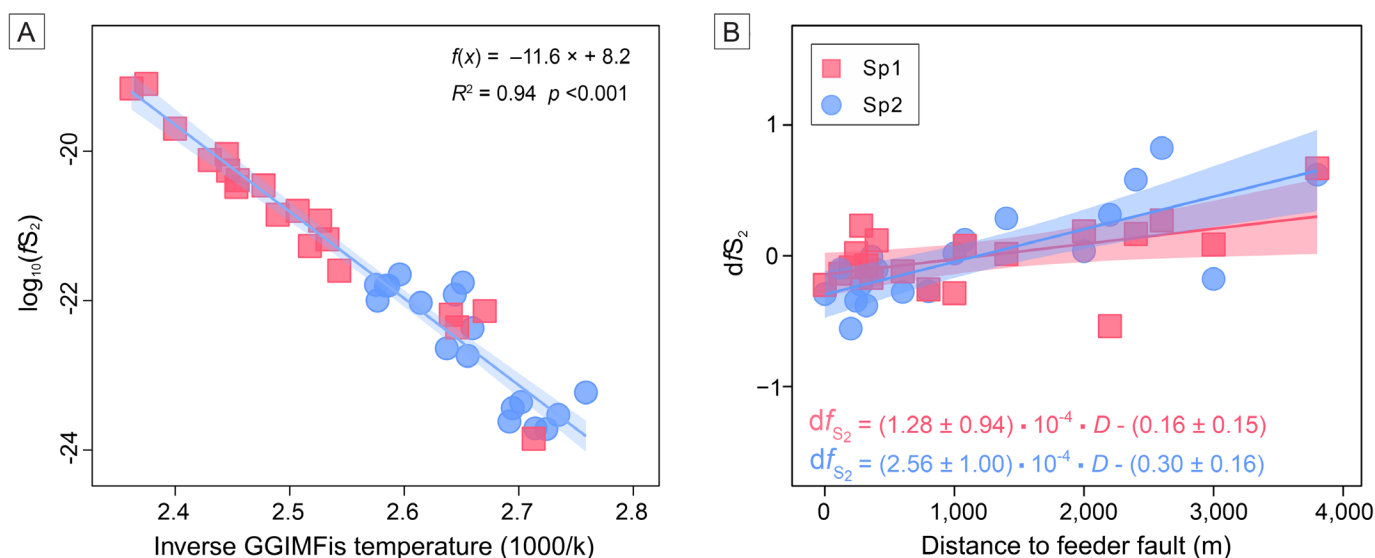


Fig. 10. Sulfur fugacity-temperature relationship and spatial variation of residual sulfur fugacity. (A) $\log_{10}(f_{\text{S}_2})$ versus inverse GGIMFis temperature for sphalerite samples in each generation. (B) Residual sulfur fugacity (df_{S_2}) values versus distance to the feeder fault. Abbreviation: GGIMFis = Ga, Ge, In, Mn, and Fe in sphalerite.

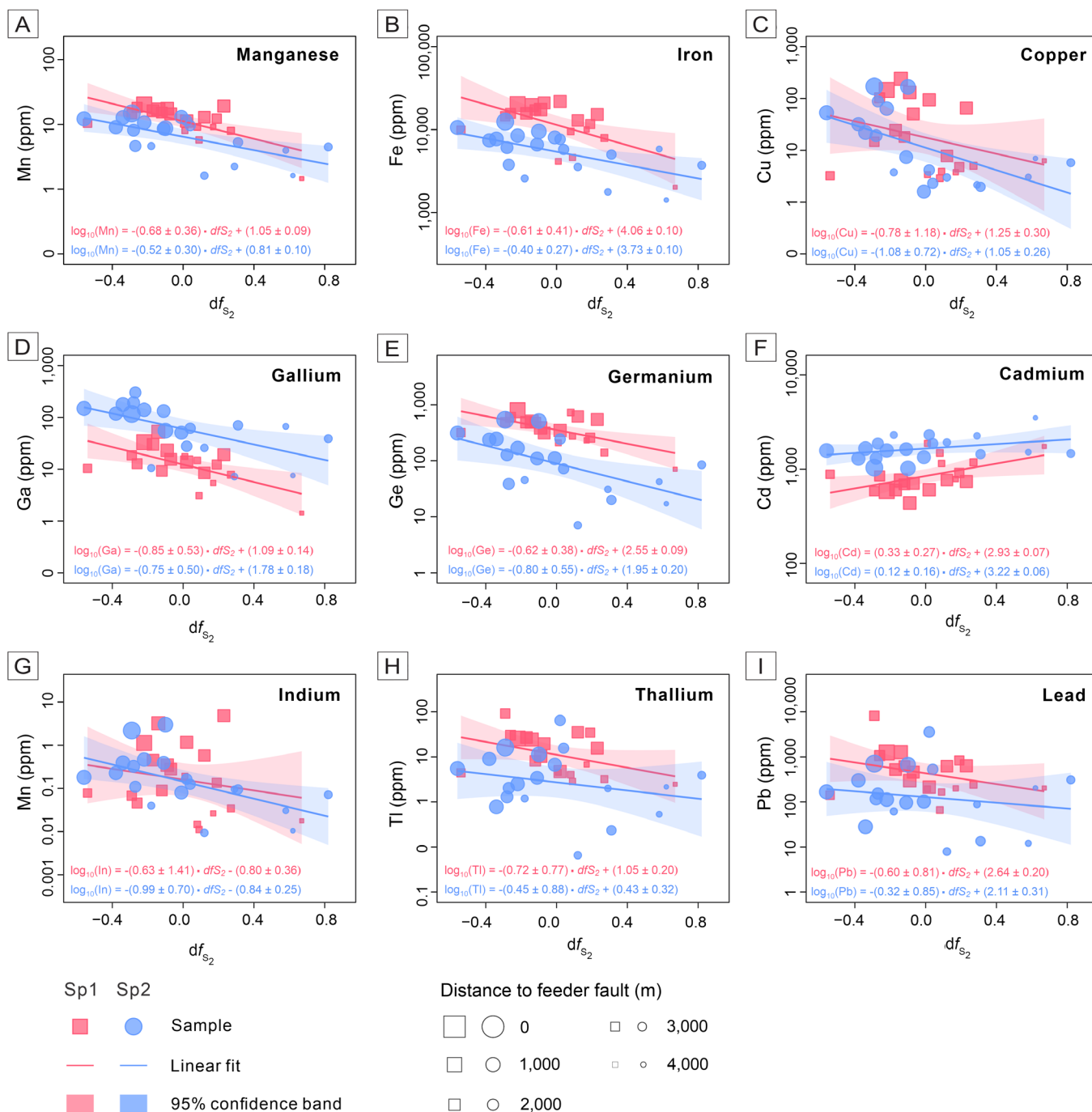


Fig. 11. Trace element contents in sphalerite from Nayongzhi as a function of df_{S_2} . (A) Mn. (B) Fe. (C) Cu. (D) Ga. (E) Ge. (F) Cd. (G) In. (H) Tl. (I) Pb. Symbol size represents the relative distance to the feeder fault. The plots also include the fitting lines and corresponding 95% confidence bands for each generation.

To evaluate the degree to which these generational differences can be explained by temperature changes, we quantitatively compared the observed mean differences with temperature-dependent variations expected from global trends (Frenzel et al., 2016). We first calculated the expected variations in trace element contents, $\Delta \log_{10}(c_i)$, by multiplying the observed temperature difference ($\Delta T = -24.0 \pm 3.4^\circ\text{C}$) with the temperature-dependent slopes of different elements,

C_{Temp} , reported in Frenzel et al. (2016). The expected variations were then compared with the observed difference from the fitted linear models (eq. 2). If the magnitude and signs of these differences match within error, it suggests that the temperature difference in the fluids that formed Sp1 and Sp2 can account for the observed trace element variations.

The comparison results for these elements are summarized in the first five columns of Table 5. The results clearly indi-

Table 5. Temperature Effect on Temporal and Spatial Variations of Trace Elements in Nayongzhi Sphalerite

Variables	C_{Temp}	Generations			Distance		
		ΔT_{Gen}	$\Delta \log_{10}(C_i)_{Gen}$	$C_{Temp} \cdot \Delta T_{Gen}$	ΔT_{Dist}	$\Delta \log_{10}(C_i)_{Dist}$	$C_{Temp} \cdot \Delta T_{Dist}$
Mn	0.051 ± 0.009	-24.0 ± 3.4	-0.24 ± 0.05	-1.22 ± 0.28	-27.6 ± 6.2	-0.87 ± 0.08	-1.41 ± 0.40
Fe	0.014 ± 0.004	-24.0 ± 3.4	-0.33 ± 0.04	-0.34 ± 0.11	-27.6 ± 6.2	-0.83 ± 0.07	-0.39 ± 0.14
Cu	0	-24.0 ± 3.4	0	0	-27.6 ± 6.2	-1.62 ± 0.30	0
Ga	-0.032 ± 0.014	-24.0 ± 3.4	0.68 ± 0.09	0.77 ± 0.35	-27.6 ± 6.2	-1.13 ± 0.17	0.89 ± 0.43
Ge	-0.044 ± 0.009	-24.0 ± 3.4	-0.60 ± 0.09	1.06 ± 0.26	-27.6 ± 6.2	-1.04 ± 0.17	1.21 ± 0.37
Ag	0	-24.0 ± 3.4	0	0	-27.6 ± 6.2	-	0
Cd	0	-24.0 ± 3.4	0.29 ± 0.03	0	-27.6 ± 6.2	0.40 ± 0.06	0
In	0.018 ± 0.018	-24.0 ± 3.4	0	-0.43 ± 0.44	-27.6 ± 6.2	-2.01 ± 0.26	-0.50 ± 0.51
Tl	n.a.	-24.0 ± 3.4	-0.61 ± 0.17	n.a.	-27.6 ± 6.2	-1.06 ± 0.30	n.a.
Pb	n.a.	-24.0 ± 3.4	-0.53 ± 0.17	n.a.	-27.6 ± 6.2	-0.84 ± 0.31	n.a.
$\log_{10}(f_{S_2})$	-	-24.0 ± 3.4	-1.85 ± 0.33	-	-27.6 ± 6.2	-1.39 ± 0.59	-
df_{S_2}	-	-24.0 ± 3.4	-	-	-27.6 ± 6.2	0.73 ± 0.15	-

Notes: C_{Temp} denotes simple linear model coefficients for specific elements with temperature from Frenzel et al. (2016); ΔT_{Gen} denotes the average temperature change between different generations; $\Delta \log_{10}(C_i)_{Gen}$ denotes average differences in trace element concentrations between generations; $C_{Temp} \cdot \Delta T_{Gen}$ denotes expected differences in trace element concentrations between generations if differences were mostly caused by temperature; ΔT_{Gen} denotes the average temperature change from feeder fault to the barren dolostone (~4 km); $\Delta \log_{10}(C_i)_{Dist}$ denotes the average changes in trace element concentrations at a distance of ~4 km; $C_{Temp} \cdot \Delta T_{Dist}$ denotes the expected changes in trace element concentrations with distance if temperature changes were the main driver for the observed variations; n.a. indicates variables without model coefficients and thus are not quantifiable; dashes indicate no data available

cated that the variations of Fe, Cu, Ga, and Ag are similar to expectations from the global trends, whereas Mn shows a much larger difference between the two generations than expected. On the other hand, Ge behaves exactly in the opposite way from expectations, and Cd and In show a difference where none would have been expected. For Tl and Pb contents, no reference data is available to evaluate the expected temperature effect.

These results indicate that temperature differences could explain the observed temporal variations of Fe, Cu, Ga, Ag, and, to some degree, Mn. However, for Ge, Cd, In, and large parts of the observed variations in Mn contents, other factors, rather than temperature changes, may cause the observed temporal variations. The most plausible explanation for this would be a change in the composition of the deep mineralizing fluid.

Figure 12 clearly shows that the trace element contents of the two generations show different correlation trends with formation temperatures. For instance, Ga shows a negative correlation with temperature in Sp1 and a positive correlation in Sp2. This indicates that the two sphalerite generations may reflect two separate events of sulfide mineralization rather than a continuously evolving hydrothermal process (Wei et al., 2021a). The variations in the deep mineralizing fluids may be caused by sourcing of the fluid from different fluid reservoirs at depth, which interacted with different lithologies for different durations or under different conditions and thus acquired different trace element signatures and initial temperatures. Lead isotope evidence supports this interpretation, with the earlier stage showing a slightly wider range of radiogenic Pb isotope ratios ($^{206}\text{Pb}/^{204}\text{Pb} = 17.82\text{--}17.91$; $^{207}\text{Pb}/^{204}\text{Pb} = 15.63\text{--}15.67$) than the later stage ($^{206}\text{Pb}/^{204}\text{Pb} = 17.85\text{--}18.90$; $^{207}\text{Pb}/^{204}\text{Pb} = 15.63\text{--}15.66$) (Zhou et al., 2018; Wei et al., 2024a). Similar changes in metal sources and mineralizing fluid compositions across different stages have also been proposed for the San Vicente MVT deposit, Peru (Chelle-Michou et al., 2023).

Spatial variations: Nayongzhi sphalerite exhibits considerable spatial variations in trace element contents. As distance

from the feeder fault increases, the contents of Mn, Fe, Cu, Ga, Ge, and In in sphalerite decrease, whereas Cd contents generally increase (Figs. 6, 7; App. Fig. A3). In addition, temperature and df_{S_2} are strongly correlated with distance to the feeder fault ($p < 2 \times 10^{-3}$; Table 2). Therefore, we need to reassess to what extent these two factors may exert considerable effects on the observed trends, using an approach similar to that applied to the earlier temporal variations. In this case, we compared the observed changes over a distance of 4 km from the feeder faults with the expected temperature-driven variations over the same distance.

The results in Table 5 indicate that temperature changes ($\Delta T = -27.6 \pm 6.2^\circ\text{C}$) with distance can account for the total variation in Mn contents and some of the observed variations of Fe and In contents in sphalerite. Silver contents do not exhibit statistically significant changes with distance, nor would any changes be expected. In contrast, the contents of Ga and Ge show the exact opposite behavior from the expected trends. Cadmium increases with distance whereas Cu decreases, but no changes in the contents of these elements would be expected from the temperature changes alone. Overall, these results indicate that the observed variations in the contents of Cu, Ga, Ge, Cd, and most of the variations in Fe and In are probably related to factors other than temperature change.

Figure 11 shows that most of the spatial trends in trace elements appear to correlate with spatial changes in df_{S_2} . As df_{S_2} increases, the contents of Mn, Fe, Ga, Ge, and possibly Cu, In, and Tl decrease, whereas Cd contents increase. With the exception of Fe (Scott and Barnes, 1971; Scott and Kissin, 1973; Lusk and Calder, 2004), the mechanisms by which f_{S_2} controls the incorporation of these elements into sphalerite remain poorly understood. In a previous empirical study of the Plaka deposit in Greece, Frenzel et al. (2022) reported the correlations between df_{S_2} and Co, Mn, Ge, In, Sb, and Hg contents in sphalerite and proposed reaction mechanisms to explain the observed behaviors. Manganese and Ge have negative correlations with df_{S_2} as reported by Frenzel et al. (2022), consistent with the trends observed in Nayongzhi

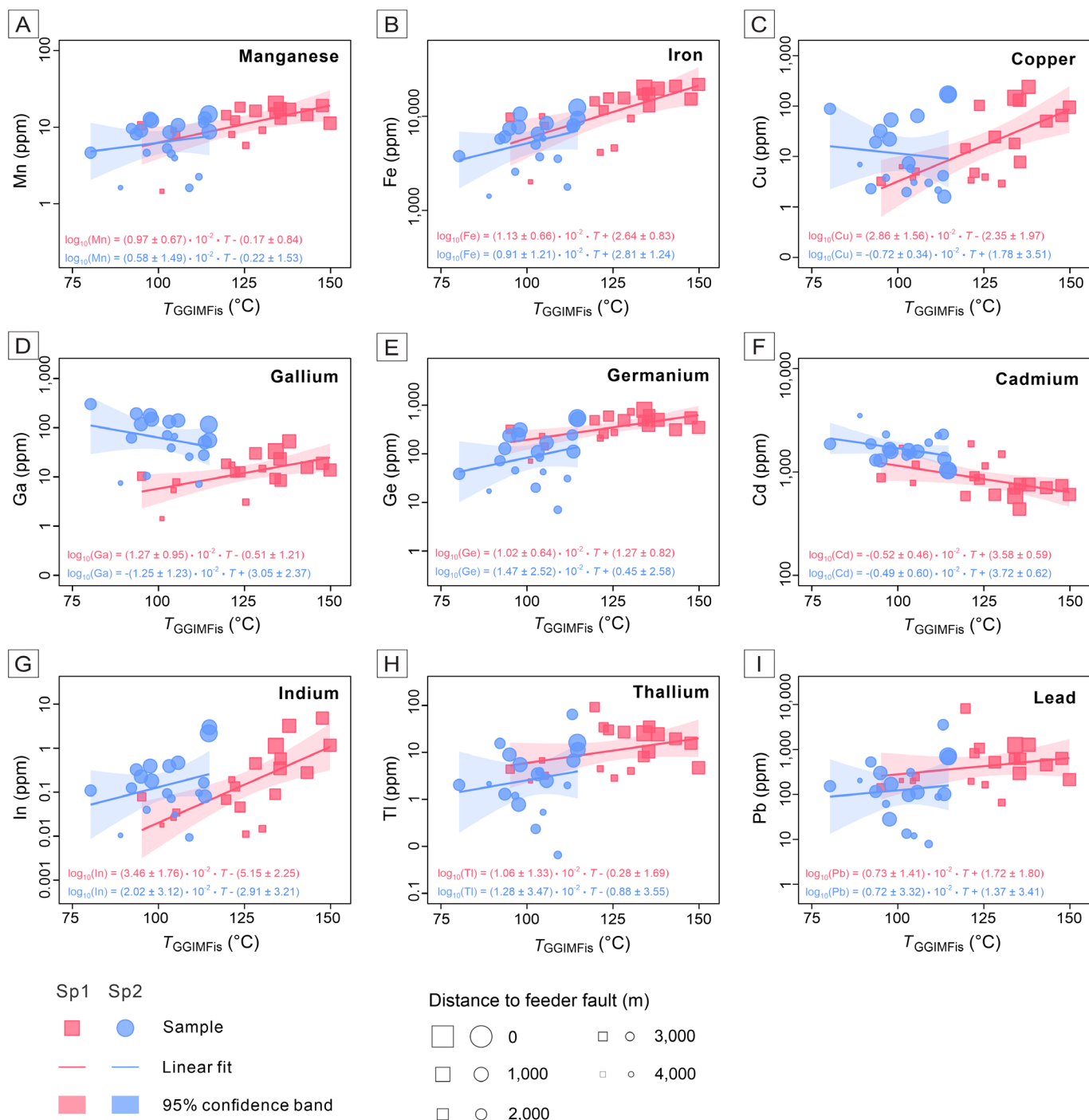


Fig. 12. Trace element contents in sphalerite from Nayongzhi as a function of GGIMFis temperature. (A) Mn. (B) Fe. (C) Cu. (D) Ga. (E) Ge. (F) Cd. (G) In. (H) Tl. (I) Pb. Symbol size represents the relative distance to the feeder fault. The plots also include the fitting lines and corresponding 95% confidence bands for each generation. Abbreviation: GGIMFis = Ga, Ge, In, Mn, and Fe in sphalerite.

sphalerite, but indium displays the opposite behavior. The present results, similar to the previous study, are solely based on correlations, which cannot allow for strong inferences to be drawn. It is worth noting that the similarities in Mn and Ge behavior may point toward similar control mechanisms behind the observed dependencies.

Several other mechanisms, in addition to df_{S_2} , may explain the observed variations to some extent. This includes trace element fractionation and coupled substitution, which will be discussed in more detail below. Additional factors, such as changes in f_{O_2} , pH, and fluid salinity across the deposit, cannot be investigated thoroughly or are considered relatively unimportant.

The fractionation of elements happens when the composition of the precipitated sulfides, expressed in element ratios, differs systematically from that of those in the fluid (cf. McIntire, 1963); that is, when

$$K_{XY} = \frac{(C_X^{\text{sulfide}})/(C_Y^{\text{sulfide}})}{(C_X^{\text{fluid}})/(C_Y^{\text{fluid}})} \neq 1, \quad (4)$$

where X and Y denote two generic elements. K_{XY} denotes the (empirical) equilibrium partitioning constant for the partitioning of X into the sulfides relative to Y . In the context of sphalerite, where most of the investigated trace elements substitute for Zn, X should be a generic trace element and Y should be Zn (cf. Frenzel, 2023). If K_{XZn} is systematically <1 , then the fluid would gradually be enriched in the trace element X relative to Zn with increasing distance from the feeder zones. This should, in turn, cause an increase in the contents of X in sphalerite. For instance, the preferential precipitation of galena relative to sphalerite at Nayongzhi is the most plausible explanation for the observed increase in Zn/Pb ratios across the deposit. The accompanying change in the fluid composition likely accounts for the observed variations of Pb contents in sphalerite, assuming that Pb incorporation is related to fluid composition. Other trace elements may be governed by similar mechanisms. If this was indeed the dominant mechanism, we may infer based on the observed trends that virtually all of the investigated trace elements showing spatial variations are preferentially incorporated into the precipitated sulfides, as indicated by decreasing concentrations with distance. The residual ore fluids appear to be enriched in Cd relative to Zn.

Coupled substitution may also account for some of the observed variations in the contents of nondivalent elements. The trivalent (Ga^{3+} and In^{3+}) and tetravalent elements (e.g., Ge^{4+}) are incorporated into the sphalerite lattice through coupling with monovalent cations (e.g., Cu^+ and Ag^+) or vacancies for charge compensation (Johan, 1988; Belissont et al., 2016; Li et al., 2020a; Torró et al., 2023, and references therein), though some studies report related nanominerals in colloform or postmetamorphic recrystallized sphalerite (e.g., Cugerone et al., 2020; Sun et al., 2024).

At Nayongzhi, transmission electron microscope studies revealed the incorporation of these elements into the sphalerite lattice (Wei et al., 2021a). The observed correlations between Cu and ($\text{Ga} + \text{Ge} + \text{In}$) ($R^2 = 0.33$; $p < 2.2 \times 10^{-22}$) support the proposed coupled substitutions (Fig. 13), though deviations from the 1:1 line may involve additional mechanisms, such as deficiencies in other charge-compensating elements (e.g., halogens) or the presence of vacancies (e.g., Cook et al., 2015; Frenzel et al., 2020). More importantly, the decrease in Cu contents with distance from the feeder faults may, to some extent, drive the observed decreases in Ga, Ge, and In, as observed in In-Ga-Cu spatial coupling in the Morococha district, Peru (Benites et al., 2021). Thus, the coupled substitutions may contribute to the observed spatial variations in the contents of trace elements, potentially driven by fractionation effects for some elements within the substitution pairs such as Cu.

In summary, the observed variations in trace element contents cannot be attributed to a single major factor. Instead, several factors likely combine to produce the observed trends,

including a decrease in temperature, an increase in dfs_2 , and an increasing degree of trace element fractionation in the fluid with increasing distance from the feeder fault of the deposit.

A sphalerite proximator?

In addition to the implications for genetic processes, the recognition of spatial variations in sphalerite composition may aid in mineral exploration in the Nayongzhi district. To achieve this goal, we used multivariate linear regressions to develop a reliable distance estimator—the sphalerite proximator.

To enhance the practicality of the proximator and reduce reliance on the petrological study, the LA-ICP-MS data set from Nayongzhi sphalerite used in equation (2) was first reprocessed to minimize the generation effects. A stepwise multivariate linear regression was then used to identify the most relevant elements suitable as indicators for the distance of a sample to the fluid influx center. The results of the stepwise fitting procedure, including R^2 values and p values for individual models, are summarized in Table 6. Although strong collinearities exist among some of the trace elements in sphalerite, the fitting procedure identified the combination of Mn, In, and Ga as the most relevant for distance estimation. The overall model yielded an excellent fit, with an overall adjusted R^2 value of 0.83 (Table 7). The final model can be expressed by the following formula:

$$\text{Distance} = \log_{10} (C_{\text{Mn}}^{-1.87} \times C_{\text{In}}^{-0.52} \times C_{\text{Ga}}^{-0.63}) + (3.42 \pm 0.48) \quad (5)$$

where *Distance* is the distance to the feeder fault in kilometers (km), $\log_{10}$ represents the logarithm (base 10), and the contents of the trace elements (Mn, In, and Ga) in sphalerite are all expressed in units of parts per million.

Figure 14A reveals that, as expected, the predicted distances to the feeder fault at Nayongzhi based on the three

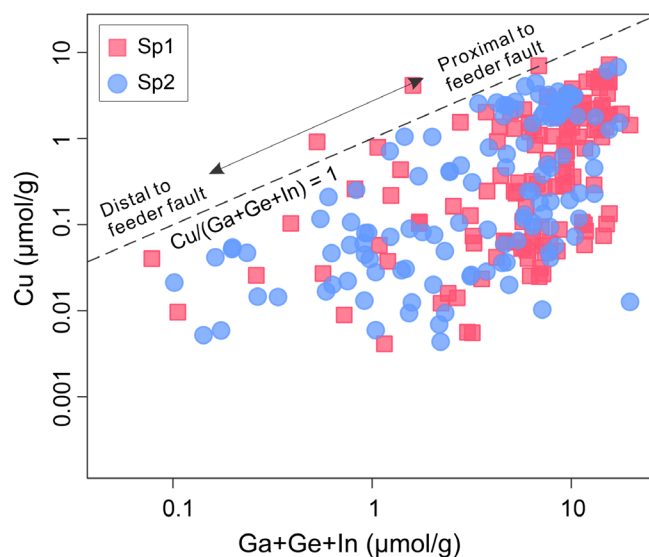


Fig. 13. Correlation between ($\text{Ga} + \text{Ge} + \text{In}$) and Cu in Nayongzhi sphalerite, based on individual laser ablation-inductively coupled plasma-mass spectrometry spot analyses. The data show a trend roughly subparallel to $[(\text{Ga} + \text{Ge} + \text{In})/\text{Cu}]_{\text{mol}} = 1$ line. The derivation of 1:1 line in some cases may indicate additional mechanisms influencing element distribution (see text for explanation). Abbreviations: Sp = sphalerite.

Table 6. Summary of the Forward Stepwise Multiple Linear Regression of Input Variables as a Function of Distance to the Feeder Fault

	Step 1	Step 2	Step 3	Step 4
Variables	$R^2 \times$	$p \text{ Mn} + X$	$p \text{ Mn} + \text{In} + X$	$p \text{ Mn} + \text{In} + \text{Ga} + X$
+ Mn	0.62	-	-	-
+ Fe	0.54	0.53	0.54	0.06
+ Cu	0.43	1.4×10^{-3}	0.15	0.15
+ Ga	0.32	1.2×10^{-5}	2.9×10^{-4}	-
+ Ge	0.31	0.06	0.02	0.55
+ Cd	0.27	0.66	0.57	0.11
+ In	0.61	6.2×10^{-5}	-	-
+ Tl	0.18	0.17	0.08	0.89
+ Pb	0.12	0.43	0.19	0.82

Notes: R^2 denotes the coefficient of determination between variables and distance; p values provide the indicators of statistical significance—the smaller the p value, the higher probability of a relationship existing with the respective explanatory variable; X represents one of the variables in this table; bold values indicate the smallest p value at each step of this linear regression analysis; dashes indicate no data available

elements correlate well with those of observed values along the line with a slope of 1/1. The application of the sphalerite proximator equation (eq. 5) accurately identifies the sampling locations for the two known sample suites (Fig. 14A).

Overall, the sphalerite proximator appears to enable effective predictions of the distance to the fluid influx center of the deposit. To efficiently vector the fluid center, it is necessary to combine distance estimates from multiple samples collected from different locations. Assuming isotropic conditions, circles representing the estimated distance from each sample can be plotted in two-dimensional space (Fig. 14B). The feeder zone is inferred from the high-density overlap of circle intersections and is well estimated by the present sample set at Nayongzhi, as expected.

A cautionary note is warranted regarding the application of the proximator. Systematic differences between sphalerite generations within the sample set can introduce biases into

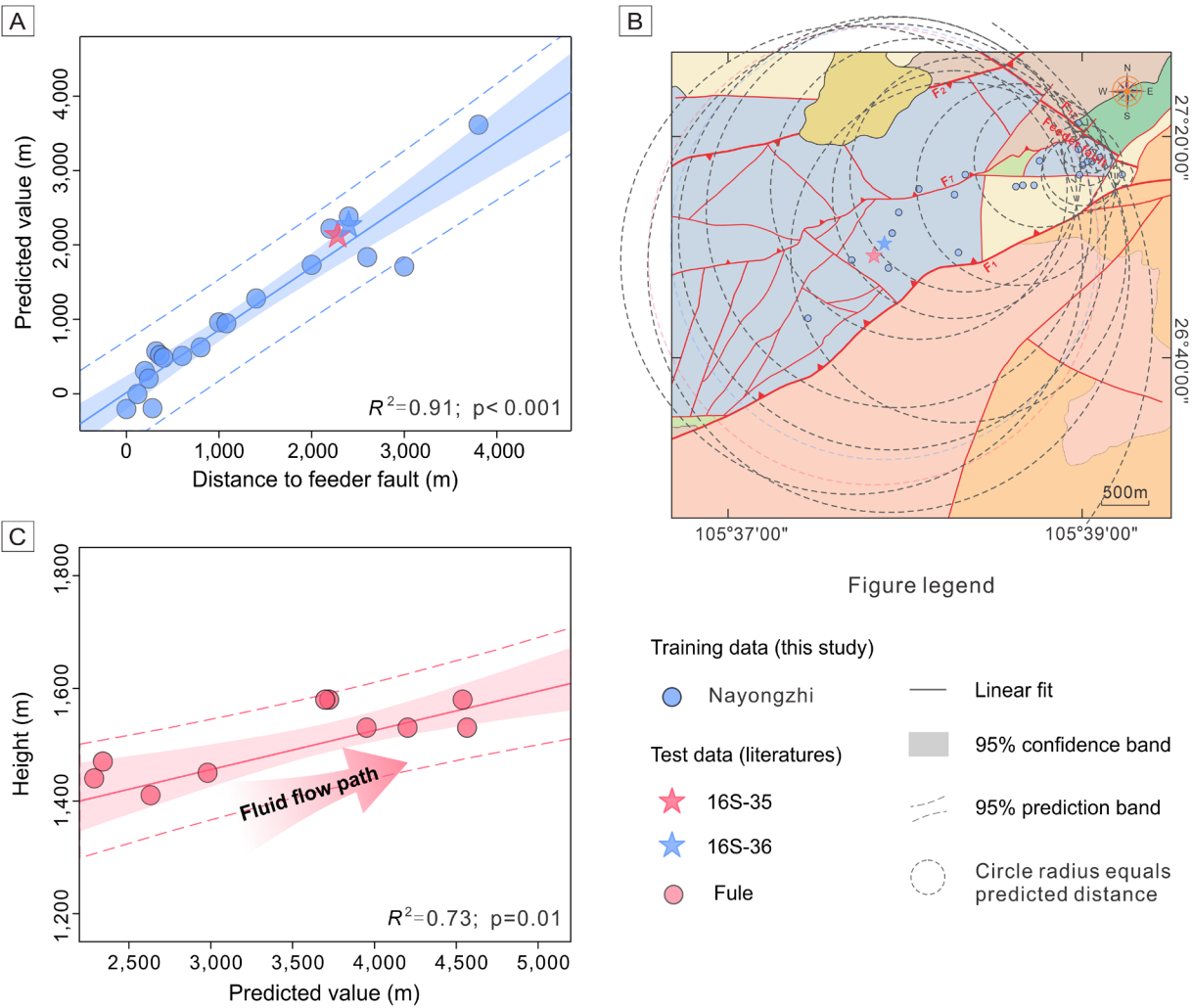


Fig. 14. Application of the sphalerite proximator to predict distance to the fluid flux center. (A) Comparison between observed and predicted distance to the feeder fault using the sphalerite proximator (eq. 5; see Table 7). Test sample locations and laser ablation-inductively coupled plasma-mass spectrometry (LA-ICP-MS) data are from Wei et al. (2021a). (B) Spatial visualization of calculated distance to the feeder fault in the Nayongzhi deposit. Each circle represents a sphalerite sample, with the radius corresponding to the predicted distance to the feeder fault. (C) Comparison of observed and predicted distance to the fluid center in the Fule deposit, using the sphalerite proximator and LA-ICP-MS data from Wei et al. (2024b).

Table 7. Fitted Model Coefficients for Trace Elements Mn, Ga, and In as a Function of Distance to the Feeder Fault

Predictors	Estimates	Standard error	<i>p</i>
Intercept	3.42	0.48	3.2×10^{-8}
Mn	-1.87	0.32	1.5×10^{-6}
Ga	-0.63	0.16	2.9×10^{-4}
In	-0.52	0.15	1.4×10^{-3}
		<i>R</i> ²	0.83
		<i>p</i>	9.2×10^{-14}

Notes: The best fits of model coefficients for distance prediction from sphalerite compositions; estimates represent coefficients that are for log-transformed trace element contents; standard error denotes standard errors of the estimates; the *p* value indicates statistical significance; the fit model ended when the *p* value > 0.05 (cf. Dunn, 1961)

distance estimates, particularly when only a single generation is present. Ideally, the presence of multiple generations should be determined through detailed petrographic analysis. In the absence of such analysis, the estimated distances are expected to be more uncertain than in the example presented here. This may limit the usefulness of the proximator in cases where sphalerite generations are spatially segregated or exhibit a regular pattern that could introduce systematic biases. Although this is not the case at Nayongzhi, such scenarios may occur in other deposits, as discussed in the following section.

Implications for mineral exploration

MVT mineralizations often have close structural associations with extensional faults at the deposit or district scale (Leach et al., 2005; Paradis et al., 2007). Metal zonation patterns in the deposits commonly result from the influx of warm, metal-rich fluids through these faults and mixing with cooler fluids or reaction with the surrounding rocks (Wilkinson, 2014). The recognition of deposit-scale zonation patterns not only helps to predict fluid flow paths and identify the locations of the main feeder structures but also aids in targeting new orebody locations. The zonation patterns thus provide an effective tool for near-mine exploration. In many cases, however, zonation patterns in MVT deposits are not clearly identified, probably due to their sometimes relatively subtle nature and to structural complexities within individual deposits.

The sphalerite proximator presented in this study provides an additional and effective method for vectoring fluid flow paths and targeting new orebodies by detecting subtle, low-level elemental geochemical anomalies preserved in sphalerite, where other indicators of zonation are absent. For this approach to be effective in MVT ore exploration, its applicability across different deposits must be validated. We thus applied equation (5) to the structurally controlled Fule Zn-Pb deposit in South China, based on the sphalerite mineral chemistry data set of Wei et al. (2024b). The results correctly identify fluid flow paths and the relative distance to the feeder structure, although the latter likely involves considerable uncertainty (Fig. 14C). This uncertainty may stem from the sampling and the absence of a petrologic study. As mentioned in the previous section, enhanced sampling density and particularly more detailed petrographic observations may help decrease prediction uncertainty. Nevertheless, the proximator remains effective even without prior petrologic examinations in this

case, and the predictions suggest significant deep exploration potential in the Fule area, supported by recent discoveries of deep orebodies (Li et al., 2020b). This example suggests that our proximator may be more broadly applicable to other deposits and potentially useful for targeting new ore at depth.

The main caution to the proximator's wider application is the potential variability of zonation patterns across different MVT deposits due to differences in the ore-forming processes. This variability may cause the observed trends to disappear or reverse, rendering the use of the proximator ineffective. Major elements do not always show the same zonation pattern. For instance, in contrast to Nayongzhi, Zn/Pb ratios in the MVT deposits of the Viburnum trend, United States, decrease with increasing distance from the fluid influx center (cf. Cavender et al., 2016). Similar differences may occur for some of the trace elements. Furthermore, even when the proximator is applicable, it may not accurately predict absolute distances due to differences in fluid flow velocities, host-rock permeabilities, and the resulting geochemical gradients across individual deposits. In such cases, it should be usable as an indicator for the relative distances of different samples to the fluid influx center, which is sufficient for vectoring. Despite these reservations, we would expect the proximator to be largely applicable in the Sichuan-Yunnan-Guizhou triangle region in South China, particularly in the deposits around Nayongzhi. This is because the region hosts abundant MVT Zn-Pb deposits that are closely associated with extensional faults, which are considered the main conduits for deep fluid migration (Zhang et al., 2015; Han et al., 2023). In addition, the fluid compositions and ore precipitation mechanisms show greater similarities to those of the Nayongzhi deposit than for other deposits around the world.

In addition, the general concept of sphalerite proximator models may have broader applications in different genetic deposit types, including volcanogenic massive sulfide, clastic-dominated, skarn, and epithermal vein systems, where sphalerite is also an important sulfide phase (e.g., Leach et al., 2005; Meinert et al., 2005; Simmons et al., 2005; Huston et al., 2010). Many of these deposits also exhibit well-defined zoning patterns in terms of the major metals, mineral assemblages, and ore textures. Targeted investigations are necessary to extend the usefulness of sphalerite in these contexts. LA-ICP-MS analyses are becoming increasingly accessible in terms of prices and availability in commercial laboratories, research centers, and universities worldwide (Cooke et al., 2020; Sader and Ryan, 2020). This may enable the development and use of more robust sphalerite vectoring tools and help to increase the effectiveness of exploration and decrease the number of drill cores required for discovery.

Conclusions

The present study showed that sphalerite trace element signatures and carbonate C-O isotope compositions record important zonation patterns at the Nayongzhi MVT deposit, South China. Our most important conclusions from this study are as follows:

1. Carbonate-oxygen isotopes of ore-related dolomite show increasing $\delta^{18}\text{O}$ and $\delta^{13}\text{C}$ values with distance from the feeder fault. This zonation pattern is dominantly controlled

- by the intensity of fluid-rock interaction between the ore fluids and the host dolostones.
- The trace element contents of sphalerite exhibit systematic spatiotemporal variations. Temporally, early sphalerite is relatively enriched in Mn, Fe, Ge, and Tl but depleted in Ga and Cd. These differences are most probably governed by differences in fluid temperatures and compositions between the two major influx events. Spatially, sphalerite near the feeder fault contains higher levels of Mn, Fe, Cu, Ga, Ge, and In, which decrease with distance, whereas Cd exhibits the opposite behavior. The exact reasons for these variations remain unclear, but they are probably mostly due to variations in df_{S_2} and fluid temperature as well as metal fractionation away from the feeder fault.
 - Data analysis by multiple linear regression models revealed that spatial variations in Mn, Ga, and In contents in sphalerite can be combined in a sphalerite proximator to vector toward the fluid flux center and identify fluid pathways at Nayongzhi. Given adequate sampling density and detailed petrographic examinations, the sphalerite proximator could be a new robust vectoring tool to identify fluid flow paths and aid in near-deposit exploration in the Sichuan-Yunnan-Guizhou triangle region and potentially elsewhere.

Acknowledgments

This research was financially supported by the National Natural Science Foundation of China (92162218, 42302101, and 42173025); Major Collaborative Innovation Projects for Prospecting Breakthrough Strategic Action of Guizhou Province ([2022]ZD004); and the Sino-German Postdoc Scholarship Program 2022 (57607866). The authors would like to thank the geologists of the Derong Mining Group Co. Ltd. for their help during field work and for fruitful discussions on the structural framework and metal distributions at the deposit. We are also grateful to Ivan Belousov, Paul Olin, Sarah Gilbert, Jing Gu, and Zhihui Dai for their assistance with LA-ICP-MS trace element and C-O isotope analyses. Finally, the authors would like to thank Lisard Torró and an anonymous reviewer for their constructive feedback and David Cooke for editorial handling.

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